

Online Appendix to

“Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS”

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A. Data Reconstruction and Treatment Contract

A.1. Source files and sample flow

The stock analysis uses two restricted IPUMS CPS extracts. The wide file supplies January 2017–July 2026 records; a second file supplies the five March Basic Monthly samples that replace incorrectly selected March ASEC samples in 2017–2021. The replication manifest records extract identifiers, creation metadata where available, byte sizes, and SHA-256 hashes without redistributing microdata. The March replacement occurs before eligibility filters or occupation routing. The superseded rows contribute no positive-final-weight analysis records; the replacement adds 252,862 eligible records.

The wide file is IPUMS extract 9: its request was submitted on August 25, 2026, and the completed-file receipt was recorded four minutes later. Separate hashes authenticate its submitted specification, data, DDI, and basic codebook. The March repair is extract 11; its actual request, returned API response, data, DDI, and basic-codebook hashes are verified, but its surviving API response contains no submission or completion timestamp, so none is inferred from filesystem metadata. Flow models additionally use extract 10 for LNKFW1MWT; it was recorded complete when checked on August 31, 2026, but the check time is not called an API completion time. Exact timestamps and hashes appear in the input manifest and provenance receipt. Restricted files, account metadata, download links, and direct identifiers are not redistributed.

TABLE 1. Calendar and cell reconstruction audit

Object	Count	Audit rule
Expected calendar months	115	January 2017–July 2026
Observed months	114	October 2025 not collected
Static months	113	December 2022 transition excluded
Primary occupations	468	named fixed support
Complete occupation–month grid	52,884	468×113
Positive-total fitted cells	51,891	includes one-sided zeros
Both-age-zero cells	993	no likelihood term
One-sided zero cells retained	12,965	finite boundary contributions

Notes: October 2025 is absent, not coded zero or interpolated. December 2022 is observed but excluded from the static pre/post fit.

Eligibility requires age 18–65, employment, a valid occupation, and positive WTFINL. The analysis scans 9,843,021 source rows, retains 5,322,047 eligible source records, and cre-

ates 6,188,956 route-expanded descendants. The last count is fractional before 2020 and is not a count of distinct respondents. The principal age groups are formed after routing. Of the complete grid, 13,883 young cells and 1,068 older cells have zero respondent-equivalent records. These sparse-cell facts motivate the finite-sample audit; they do not justify selecting cells on realized young counts.

A.2. Reconstructed preperiod treatment

The substantive treatment is rebuilt using only January 2017–November 2022 stock. On the fixed 468-occupation support, the employment-weighted beta mean is 0.37520 and standard deviation is 0.21041. The tie-preserving cut values are 0.153846, 0.324615, 0.456522, and 0.537037. Webb software exposure is standardized over the same 71-month construction window. Table 2 compares the rebuilt contract with the historical production contract.

TABLE 2. Historical and reconstructed treatment contracts

Contract	Construction window	Post stock used	Q5–Q1 coefficient
Historical production	full static panel	Yes	–0.134554
Reconstructed	Jan. 2017–Nov. 2022	No	–0.132109

Notes: Both coefficients use the corrected 113-month outcome calendar and identical occupation support. Nine occupations change quintile. The reconstructed-minus-historical paired difference is 0.002445 with 95-percent interval [–0.001269, 0.006158].

The reconstructed contract contains 129, 99, 82, 67, and 91 occupations in Q1–Q5. Corresponding preperiod stock shares are 20.06, 19.96, 20.16, 20.31, and 19.51 percent. The complete membership file identifies all nine occupations whose historical and rebuilt labels differ. All revised substantive models use the reconstructed labels unless explicitly described as a historical audit.

A.3. Survey-production dates

Official BLS documentation establishes population-control discontinuities in January 2025 and January 2026. BLS did not revise the preceding official months onto the new bases and does not publish counterfactual age-by-occupation weights. October 2025 was not collected; November collection began late, used reduced overlap and a modified weighting procedure, and had a lower response rate. IPUMS later processed the reissued January 2026 file and corrections to longitudinal identifiers. The stock and flow manifests preserve their respective extract dates and hashes. Endpoint and exclusion checks diagnose

reliance on these dates, but no aggregate adjustment factor is imposed on detailed cells.

B. Occupational Mapping and Analysis Support

B.1. Bridge construction

The early panel reports Census-2010 occupations and the later panel Census-2018 occupations. Exposure inputs additionally span SOC vintages. The versioned bridge first routes 503 early source codes into 568 target codes. Fifty-six sources have more than one target, with maximum multiplicity seven. Official shares partition each source record's final weight; separate young and older routing would require validation data that the public adjacent-vintage files do not provide. The routed early and current stocks reconcile to their sources with relative error below 2×10^{-16} .

In the frozen mapping-only lookup, the equal-administrative AIOE variant has full-route coverage for 480 of the 503 pre-2020 source codes. This 480-code mapping count is not the regression sample: intersecting the rebuilt beta and Webb inputs with positive young preperiod stock yields the 468-occupation primary analysis support below.

The primary joint beta/Webb universe contains 468 occupations and 87.19 percent of stock in the audited candidate universe. Table 3 gives mutually exclusive exclusion counts. A machine-readable file lists each included and excluded occupation, its preperiod stocks, scores, and reason. This file, rather than a hard-coded count in the paper, defines support.

TABLE 3. Primary-support exclusions

Reason	Occupations
Missing Rule-A beta only	41
Missing Webb software only	22
Missing both exposure inputs	8
Nonpositive young preperiod stock	2
Total excluded	73
Primary retained	468

B.2. Separating score correction from population expansion

A literal exact-code merge between SOC-2010 AIOE and post-2018 employment data leaves only 3.33 percent of computer-and-mathematical employment. The version-aware route

reaches 97.7 percent. The four-row continuous comparison separates the two operations that an aggregate “crosswalk effect” would conflate:

The source workbook standardizes AIOE without employment weights across 774 source occupations: its mean is approximately zero and its sample standard deviation is approximately one. A raw AIOE point is therefore one unit of that source index, not a probability or percentage. As a common descriptive increment, the source-workbook interquartile difference is 1.879 raw units. The historical four-row audit below uses a different, explicitly fixed unit: one full-window employment-stock-weighted standard deviation of repaired equal-administrative AIOE on the expanded 495-occupation AIOE-plus-Webb support. The retained artifact does not store the numerical weighted standard deviation, so I do not manufacture a raw-point conversion.

TABLE 4. AIOE taxonomy repair and support expansion

Score	Support	Occupations	Coefficient per fixed SD
Literal original	original	410	−0.01885
Repaired	same original	410	−0.01920
Repaired	expanded	495	−0.03156
Repaired	expanded, excluding SOC15	—	−0.02940

Notes: The first row is an invalid vintage merge retained as an implementation audit, not a defensible specification. Row 1 to 2 isolates score repair on fixed support; row 2 to 3 changes the population; the last row tests whether readmitted computer-and-mathematical occupations alone account for that expansion movement. Every coefficient uses the fixed weighted-SD unit defined in the text; none is a percentage or a coefficient per raw AIOE point.

The invalid merge is documented as a failure encountered in this project. I do not claim that a published paper made the same error without direct verification of its code and occupational universe.

B.3. Split-route and age-allocation sensitivity

Within the primary support, split sources contribute 14.93 percent of young and 17.75 percent of older early stock. Ninety-nine target occupations can receive a split route. A one-to-one-target restriction and an aggregation to stable Census-2010 categories both retain negative historical-contract estimates, but each changes the target population. They are mapping sensitivities, not repairs of the primary bridge.

Using the same route proportions by age is transparent but unvalidated. Accounting ranges that allocate every ambiguous source in all feasible ways place the 2017–2019 young-

to-older stock ratio between 0.1031 and 0.1480 in Q1 and between 0.0790 and 0.1175 in Q5. These are stock-accounting bounds, not bounds on the nonlinear regression coefficient. An earlier historical-contract tilt involved 6.88 percent of early stock and is retained only as an implementation checkpoint; the current-contract sensitivity follows below. No adjacent coding vintage is treated as a same-person validation sample.

B.4. Current-contract route-allocation sensitivity

The current 113-month contract repeats the exercise on the 468-occupation support. Thirty-one split source occupations, accounting for 14.86 percent of supported early-period routed stock, receive jointly feasible young-versus-older high/low allocation-odds tilts $K \in \{0.25, 0.5, 2/3, 1, 1.5, 2, 4\}$. Each grid point preserves source-age-month mass and official structural route zeros. Fixed-label rows hold the no-tilt exposure groups constant; rebuilt-treatment rows recompute them after allocation. Table 5 keeps those objects separate.

TABLE 5. Current-contract bridge-allocation sensitivity

Model	Exposure labels	Coefficient range	Paired intervals excluding zero
Pooled	Fixed at $K = 1$	$[-0.1433, -0.1167]$	0 of 6
Pooled	Rebuilt at each K	$[-0.1446, -0.1163]$	0 of 6
Family by month	Fixed at $K = 1$	$[-0.0748, 0.0409]$	0 of 6
Family by month	Rebuilt at each K	$[-0.0783, 0.0439]$	5 of 6

Notes: The grid is $K \in \{0.25, 0.5, 2/3, 1, 1.5, 2, 4\}$, where K is the relative young-versus-older high/low allocation odds for jointly feasible splits. Ranges summarize coefficients, not confidence intervals. The last column counts occupation-clustered paired intervals for the six nonbaseline points relative to $K = 1$. All rows are post-outcome mapping sensitivities; the grid is neither an identified set nor a sharp bound.

All pooled paired occupation-clustered intervals relative to $K = 1$ contain zero under both label rules. For rebuilt treatment in the family-month model, however, the coefficient moves from -0.0217 at $K = 1$ to 0.0439 at $K = 0.25$; the paired movement is 0.0656 with interval $[0.0031, 0.1280]$. Five of the six rebuilt-treatment family-month comparisons exclude zero, even though rebuilding changes at most two occupations at a grid point. Thus the pooled pattern is stable over this declared grid while the family-conditioned estimand is sensitive to route allocation and resulting membership. Null intervals are non-detections, not equivalence.

A separate 25-point grid allows young and older allocation odds to vary jointly while maintaining the same mass constraints. The pooled coefficient ranges from -0.1599391

to -0.0702179 . No outcome-directed optimization followed inspection of the grid, but its extrema remain an explored feasible-grid envelope rather than sharp or global bounds.

Finally, excluding every current occupation that can receive an inbound one-to-many route leaves 369 occupations. On this clean-route support, pooled coefficients are -0.1642 ($[-0.2665, -0.0618]$) with fixed labels and -0.1468 ($[-0.2415, -0.0521]$) with rebuilt labels; family-month coefficients are -0.0871 ($[-0.2325, 0.0584]$) and -0.0524 ($[-0.1897, 0.0848]$). On a separate stable-Census-2010 support of 408 occupations, the estimate is -0.1522 ($[-0.2438, -0.0606]$). On the full 468-occupation support restricted to the 77 months from 2020 onward, the current-label estimates are -0.1181 ($[-0.1941, -0.0420]$) pooled and -0.0304 ($[-0.1716, 0.1108]$) with family-by-month effects. These three exercises change the sample or calendar as well as the labeling rule and are not a single nested robustness ladder; none validates one coding vintage against another.

The inherited optional symmetric coding-error sensitivity is not executed: no authenticated dual-coded CPS sample or validated external misclassification matrix is available, immediate occupation reversals mix true mobility with reporting and coding changes, and the route-allocation exercises above are not relabelled as a coding-error correction.

C. Broad-Family Support, Profiles, and Influence

C.1. Family-by-quintile support

Table 6 reports, for every SOC two-digit family, the number of retained occupations, preperiod stock share, exposure range, quintiles present, direct-tail overlap, and nuisance-adjusted Q5 information share. Only arts/media (SOC27), health practitioners (SOC29), health support (SOC31), and sales (SOC41) span Q1 and Q5 under the reconstructed comparison. Other families connect the Q5–Q1 regression coefficient through intermediate quintiles and the imposed common Q2–Q5 slopes.

TABLE 6. Broad-family exposure support

SOC2	Family	Occs.	Stock (%)	Beta range	Quintiles	Q1/Q5	Q5 info. (%)
11	Management	29	13.56	0.367–0.586	3 4 5	No	3.04
13	Business and Financial Operations	28	6.22	0.405–0.636	3 4 5	No	5.99

Continued on next page

Table 6 continued

SOC2	Family	Occs.	Stock (%)	Beta range	Quintiles	Q1/Q5	Q5 info. (%)
15	Computer and Mathematical	15	4.31	0.500–0.950	4 5	No	0.28
17	Architecture and Engineering	19	2.28	0.406–0.727	3 4 5	No	1.87
19	Life, Physical, and Social Science	21	1.25	0.344–0.723	3 4 5	No	0.54
21	Community and Social Service	12	1.31	0.111–0.439	1 2 3	No	0.20
23	Legal	4	1.34	0.425–0.571	3 4 5	No	0.28
25	Educational Instruction and Library	7	4.53	0.229–0.567	2 3 5	No	3.20
27	Arts, Design, Entertainment, Sports, and Media	25	1.80	0.000–0.958	1 2 3 4 5	Yes	4.86
29	Healthcare Practitioners and Technical	37	6.38	0.067–0.618	1 2 3 4 5	Yes	3.61
31	Healthcare Support	14	3.66	0.074–0.875	1 2 3 5	Yes	7.26
33	Protective Service	17	2.28	0.036–0.415	1 2 3	No	1.42
35	Food Preparation and Serving Related	10	3.27	0.000–0.511	1 2 4	No	10.84
37	Building and Grounds Cleaning and Maintenance	6	2.51	0.000–0.355	1 2 3	No	1.29
39	Personal Care and Service	17	2.51	0.015–0.400	1 2 3	No	2.97
41	Sales and Related	17	10.06	0.133–0.806	1 2 3 4 5	Yes	17.40
43	Office and Administrative Support	47	11.21	0.216–0.975	2 3 4 5	No	18.92
45	Farming, Fishing, and Forestry	5	0.15	0.043–0.292	1 2	No	0.14
47	Construction and Extraction	30	6.14	0.000–0.417	1 2 3	No	1.97
49	Installation, Maintenance, and Repair	31	3.56	0.000–0.410	1 2 3	No	2.64
51	Production	52	4.14	0.000–0.438	1 2 3	No	5.67
53	Transportation and Material Moving	25	7.52	0.031–0.413	1 2 3	No	5.59

Notes: Stock is each family’s share of preperiod employment on the 468-occupation rebuilt support. Q5 information is the family’s share of nuisance-adjusted fitted information for the family-conditioned Q5 target; it is neither a sampling-cluster share nor pre-outcome residual-exposure support. Beta ranges and memberships use the corrected preperiod treatment contract. Only Q1/Q5 rows enter the changed-population direct-tail exercise.

Table 7 makes the loss and concentration of identifying information explicit. Family-by-month conditioning retains 29.7 percent of the pooled Q5 target’s nuisance-adjusted information, while the direct-tail comparison has only 6.2 effective occupations and its five largest occupation contributions supply 77.8 percent of information. Its normal-theory MDE_{80} is 0.4576 log point. The continuous within-family companion is more diffuse and has $MDE_{80} = 0.0313$ per one within-family standard deviation, but it imposes a common slope across families.

TABLE 7. Family-support information and precision diagnostics

Target and structure	Occ.	I (millions)	I/I_{raw}	I/I_{parent}	Effective occ.	Top-five share	Normal MDE_{80}
Q5–Q1, pooled	468	26.289	0.378	1.000	43.2	0.245	0.1266
Q5–Q1, SOC2 $\times$ month	468	7.803	0.112	0.297	53.3	0.218	0.1998
Direct-tail Q5–Q1, SOC2 $\times$ month	29	0.669	0.080	0.308	6.2	0.778	0.4576
Continuous within-SOC2 slope, SOC2 $\times$ month	468	261.278	0.583	0.965	48.0	0.249	0.0313

Notes: I is nuisance-adjusted Fisher information after projecting the target score on the model’s nuisance score; I_{raw} is the raw weighted treatment sum of squares. “Parent” is the corresponding pooled model on the same target and population, so the direct-tail and continuous rows each use their own pooled parent. Because target scales and populations differ, absolute I is not comparable across all rows. Effective occupations are the inverse Herfindahl count of occupation-level information contributions; the top-five share is the sum of the five largest contributions. Fitted information and its effective count are design-concentration diagnostics: neither replaces the nominal occupation-cluster count nor measures first-stage CPS sampling precision. $MDE_{80} = 2.80\widehat{SE}$ uses the occupation-cluster standard error and is a normal-theory precision diagnostic, not a rejection result. The direct-tail MDE is 0.4576 log point; the continuous MDE is 0.0313 per one within-family standard deviation (0.1084 raw beta units).

The direct-tail population retains only occupations in the four spanning families that belong to Q1 or Q5. It contains 29 occupations and 5.03 percent of primary preperiod stock. In the SOC2-by-calendar-month model its coefficient is 0.1494, with interval $[-0.1694, 0.4682]$. This benchmark is valuable because it exposes direct overlap; it is not a preferred estimate for the original population, and the interval includes large effects of either sign.

Table 8 names every occupation supplying that comparison. The imbalance is economically visible: for example, the health-support Q5 endpoint consists only of medical transcriptionists, while its Q1 endpoint includes nursing assistants and home health aides. Naming the membership clarifies what the direct-tail estimate compares; it does not make the sparse endpoints representative of their broad families.

TABLE 8. Occupations supplying direct within-family tail support

SOC2	Tail	Census occupation	Occupation	Preperiod stock (million person-months)
27	Q1	2721	Athletes and sports competitors	1.83
27	Q1	2740	Dancers and choreographers	0.94
27	Q5	2850	Writers and authors	14.91
27	Q5	2825	Public relations specialists	8.61
27	Q5	2830	Editors	8.44
27	Q5	2861	Interpreters and translators	6.14
27	Q5	2810	News analysts, reporters, and journalists	5.41
27	Q5	2840	Technical writers	4.38
27	Q5	2751	Music directors and composers	2.74
27	Q5	2805	Broadcast announcers and radio disc jockeys	2.19
27	Q5	2862	Court reporters and simultaneous captioners	2.10
29	Q1	3423	Surgical technologists	6.77
29	Q1	3402	Paramedics	6.47
29	Q1	3422	Psychiatric technicians	3.59
29	Q5	3515	Medical records specialists	11.77
29	Q5	3030	Dietitians and nutritionists	7.71
31	Q1	3603	Nursing assistants	86.89
31	Q1	3601	Home health aides	31.36
31	Q1	3630	Massage therapists	10.70
31	Q1	3605	Orderlies and psychiatric aides	3.48
31	Q5	3646	Medical transcriptionists	2.62
41	Q1	4900	Models, demonstrators, and product promoters	2.18
41	Q5	4850	Sales representatives, wholesale and manufacturing	81.01
41	Q5	4810	Insurance sales agents	38.93
41	Q5	4840	Sales representatives of services, except advertising, insurance, financial services, and travel	36.99
41	Q5	4820	Securities, commodities, and financial services sales agents	15.73
41	Q5	4800	Advertising sales agents	11.43
41	Q5	4830	Travel agents	4.26
41	Q5	4940	Telemarketers	2.87

Notes: This is the complete 29-occupation membership of the changed-population direct-tail exercise, ordered by family, tail, and stock. Preperiod stock is the cumulative CPS-weighted employed-person stock across the 71 observed months from January 2017 through November 2022: for each occupation, the producer sums the monthly young (ages 22–25) and older (ages 26–65) employment stocks. The displayed unit is one million employed-person months, not average-month employment. The list makes the direct comparisons auditable; it is not a ranking by estimated effect or influence.

C.2. Supported edges and broader beta support

The fully interacted within-family model yields 89 supported pairwise quintile-edge functionals. All 89 estimates and their standard errors are retained in the machine-readable results. They are distinct family-specific comparisons rather than repeated national Q5–Q1 estimates. Their joint covariance has rank 50, so an 89-degree-of-freedom Wald test is invalid and is blocked in the results rather than replaced by a selected list of pointwise p -values. The associated 50-dimensional family-specific coefficient system is reported separately with rank-aware inference.

Table 9 gives a transparent summary of the supported edges. Each row averages only family-specific contrasts for which both endpoint quintiles are observed, using fixed preperiod endpoint-stock weights. Because the eligible family set changes across rows, these aggregates do not impose a common national quintile profile and cannot be read as ten estimates of one population parameter.

TABLE 9. Aggregates of supported within-family quintile contrasts

Contrast	Eligible families	Estimate	Standard error	Normal p -value
Q2–Q1	14	–0.0095	0.0368	.796
Q3–Q1	12	0.0132	0.0541	.807
Q4–Q1	4	–0.0618	0.0577	.284
Q5–Q1	4	0.1020	0.0781	.192
Q3–Q2	14	0.0702	0.0348	.044
Q4–Q2	5	–0.0033	0.0354	.926
Q5–Q2	6	0.1177	0.0498	.018
Q4–Q3	9	–0.0841	0.0458	.066
Q5–Q3	11	–0.0446	0.0965	.644
Q5–Q4	10	–0.1082	0.0485	.026

Notes: For each pair, only families containing both listed quintiles enter. Family-specific edge coefficients are aggregated using fixed pair-endpoint preperiod employment-stock weights, and the covariance retains the corresponding linear functional of the full coefficient system. The ten rows have different eligible-family populations and are not a monotone national dose-response sequence. Normal p -values are descriptive and were not used to select rows; all 89 underlying supported family-specific edges and their simultaneous intervals remain machine readable.

Table 10 asks whether sparse tails arise solely from requiring Webb software exposure. Relaxing that requirement expands the population from 468 to 490 occupations, but the added occupations represent only 1.88 percent of broader-support preperiod stock. Direct tails expand from 29 to 62 occupations and from 5.03 to 12.61 percent of stock because

SOC43 joins the four original tail-spanning families. The overlap therefore improves at the margin but remains confined to five of 23 families. Coefficients change little in this descriptive population expansion; this is not a bound and does not make the two populations equivalent.

TABLE 10. Supported tails when Webb availability is relaxed

Population and classification	Occupations	Families	Tail-spanning families	Tail occupations	Tail stock (%)	Pooled coefficient
Primary support, with Webb	468	22	4	29	5.03	-0.1321
Primary support, Webb omitted	468	22	4	29	5.03	-0.1338
Broader beta-valid support, primary cuts	490	23	5	62	12.61	-0.1344
Broader beta-valid support, recomputed cuts	490	23	5	62	12.61	-0.1348

Notes: The primary support requires both beta exposure and Webb software exposure; omitting Webb from the regression on that support does not change membership. The broader support requires beta exposure but not Webb and adds 22 occupations, equal to 1.88 percent of broader-support preperiod stock. With the primary cuts or recomputed cuts it adds office and administrative support (SOC43) as the fifth family containing both Q1 and Q5. “Tail occupations” and “tail stock” count Q1 and Q5 occupations within tail-spanning families. The coefficient change from 468 to 490 occupations changes the population and is descriptive, not a paired test. On fixed support, the Webb-omitted minus Webb-included coefficient difference is -0.0017, with paired interval [-0.0102, 0.0069]; nondetection is not equivalence.

C.3. Full exposure profile

The pooled Q2–Q5 coefficients are jointly different from zero ($\chi^2_4 = 11.1383$, asymptotic $p = .0251$, multiplier $p = .0212$), but pairwise movements do not establish a monotone dose response. After SOC2-by-calendar-month conditioning, the four coefficients are jointly indistinguishable from zero ($\chi^2_4 = 2.0708$, asymptotic $p = .7227$, multiplier $p = .7422$). The four-coefficient paired movement likewise does not reject ($\chi^2_4 = 5.4169$, asymptotic $p = .2471$, multiplier $p = .2368$), despite the single Q5 movement’s pointwise interval excluding zero. Table 11 reports pointwise and simultaneous intervals and the common-draw movement for every group.

The continuous companion demeans beta within family using preperiod employment weights and imposes a common linear slope. With SOC2-by-calendar-month conditioning, the rebuilt coefficient is -0.0025 per 0.1084 raw beta units. Its interval is [-0.0240, 0.0191] in that standardized within-family unit. The unconditioned continuous coefficient is essentially zero as well. This is a different functional form from the tail contrast.

C.4. Exact numerator, denominator, and family accounting

Table 12 separates the model-free aggregate Q5–Q1 stock-ratio movement into its young numerator and older denominator. The older tail difference reinforces the negative young

TABLE 11. Exposure profile before and after broad-family monthly paths

Group	Pooled			SOC2 $\times$ month			Paired movement
	Coef.	Pointwise 95% CI	Simultaneous 95% CI	Coef.	Pointwise 95% CI	Simultaneous 95% CI	Coef. [95% CI]
Q2	-0.0784	[-0.1546,-0.0023]	[-0.1716,0.0147]	-0.0247	[-0.1228,0.0734]	[-0.1431,0.0937]	0.0537 [-0.0051,0.1126]
Q3	-0.0500	[-0.1224,0.0225]	[-0.1383,0.0383]	0.0169	[-0.0859,0.1197]	[-0.1043,0.1381]	0.0669 [-0.0047,0.1386]
Q4	-0.0896	[-0.1578,-0.0214]	[-0.1737,-0.0055]	-0.0126	[-0.1302,0.1051]	[-0.1510,0.1259]	0.0770 [-0.0138,0.1679]
Q5	-0.1321	[-0.2206,-0.0437]	[-0.2393,-0.0249]	-0.0217	[-0.1607,0.1173]	[-0.1836,0.1403]	0.1104 [0.0107,0.2102]

Notes: Q1 is omitted. Simultaneous intervals use each model's common four-target maximum- $|t|$ critical value. To prevent the same Q5 target carrying seed-dependent pointwise intervals, the Q5 pointwise and paired cells reproduce the declared single-target canonical intervals; its simultaneous cells remain the four-target family-profile result. All rows use rebuilt labels and 9,999 common occupation draws within each declared collection.

movement but is smaller in absolute magnitude. A second exact accounting separates changes in within-family young-to-older ratios from changes in older family weights. Neither accounting is forced to equal the nonlinear regression target, and no component is interpreted as a causal percentage explained.

An outcome-free design audit explains why this accounting is not presented as two saturated single-age regressions. On the 52,884 single-age occupation-month rows, unrestricted occupation-by-month indicators span the entire row space and absorb all five requested exposure-by-post columns. An additive occupation-plus-calendar-month companion retains rank five for those columns, but no outcome-bearing companion coefficients were estimated. The table therefore reports exact stock identities under the paper's normalization, not a regression decomposition of the nonlinear grouped-binomial target.

C.5. Leave-one-family and occupation influence

Leave-one-family-out estimates are computed from the same family-conditioning program and reported with the omitted family's stock and target-information contribution. These rows answer whether a broad comparison is concentrated; they are not used to choose which families belong in the main estimate. Table 13 displays the eight largest absolute movements. Omitting office/administrative support or food preparation moves the coefficient in opposite directions by about 0.04 log point, and every paired deletion interval includes zero.

Figure 1 displays the economically relevant endpoints behind those two largest leave-family-out movements: Q5 office/administrative-support stocks and Q1 food-preparation stocks. These families do not themselves span both tails, which is precisely why their paths inform the broad between-family comparison rather than the sparse direct-tail benchmark. The selection is outcome-informed and the figure is diagnostic; it neither identifies

TABLE 12. Exact descriptive accounting for the Q5–Q1 stock-ratio change

<i>Panel A. Underlying equal-month employment stocks</i>		
Series	Preperiod	Postperiod
Q1 young, ages 22–25 (millions)	2.5902	2.9044
Q1 older, ages 26–65 (millions)	21.1411	21.1983
Q5 young, ages 22–25 (millions)	1.9562	1.9654
Q5 older, ages 26–65 (millions)	21.1303	22.2367
Q1 young-to-older ratio	0.1225	0.1370
Q5 young-to-older ratio	0.0926	0.0884
<i>Panel B. Exact Q5–Q1 changes and decomposition</i>		
Component	Log points	
Young-stock Q5–Q1 change	–0.1098	
Older-stock Q5–Q1 change	0.0483	
Young-relative change (young minus older)	–0.1581	
Within-family young-to-older ratio component	–0.1293	
Changing older-family-weight component	–0.0288	
Observed boundary-mass component	0.0000	

Notes: Panel A uses equal observed-month averages over the 71 preperiod and 42 postperiod months; it reports the four underlying tail-by-age series before forming ratios. The identity $-0.1581 = -0.1098 - 0.0483$ closes to 2.5×10^{-16} . Panel B's final three rows are an exact symmetric log-Shapley accounting of the same -0.1581 change. All 88 family–tail–period denominator cells are classified before division: 52 are positive and 36 are structural absences, with no undefined active denominator. No pseudocount is used. These are model-free descriptions, not the grouped-binomial coefficient, confidence intervals, or causal shares.

TABLE 13. Largest leave-one-family-out movements for the conditioned Q5 target

Omitted SOC2	Family	Preperiod stock (%)	Re-estimated coefficient	Movement [95% paired CI]
43	Office and Administrative Support	11.21	-0.0614	-0.0397 [-0.1131,0.0337]
35	Food Preparation and Serving Related	3.27	0.0174	0.0391 [-0.0292,0.1074]
31	Healthcare Support	3.66	-0.0534	-0.0317 [-0.0772,0.0139]
13	Business and Financial Operations	6.22	0.0099	0.0316 [-0.0006,0.0638]
39	Personal Care and Service	2.51	-0.0457	-0.0240 [-0.0496,0.0015]
25	Educational Instruction and Library	4.53	-0.0351	-0.0134 [-0.0383,0.0115]
17	Architecture and Engineering	2.28	-0.0339	-0.0123 [-0.0330,0.0085]
33	Protective Service	2.28	-0.0096	0.0121 [-0.0050,0.0291]

Notes: The parent SOC2-by-calendar-month coefficient is -0.0217 . Families are ranked by the absolute coefficient movement, so this is an outcome-informed influence diagnostic rather than a preferred deletion rule. Quintiles and the continuous centering scale remain fixed; all 22 deletions are available in the machine-readable file.

a pandemic mechanism nor supplies a deletion rule.

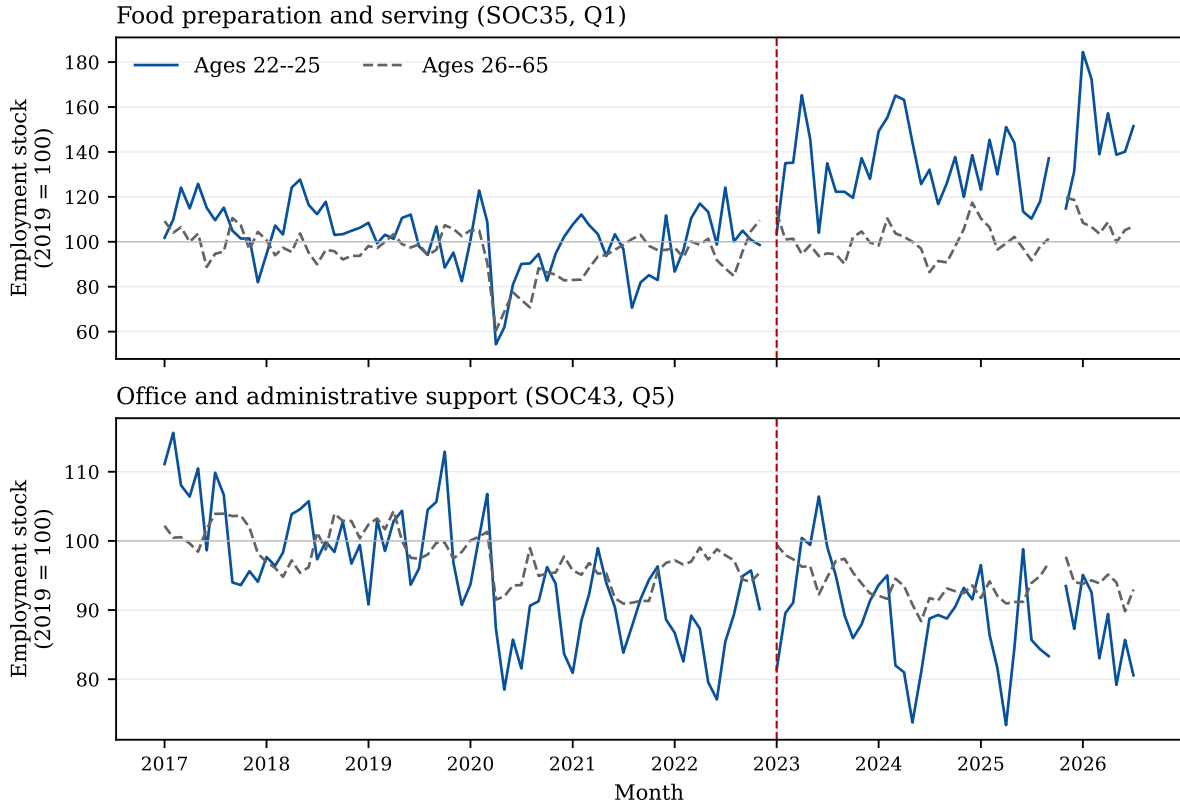


FIGURE 1. Employment paths for the two largest leave-family-out movers

Notes: The panels show Q1 food-preparation and serving occupations (SOC35) and Q5 office and administrative-support occupations (SOC43), the family-tail endpoints relevant to the broad Q5–Q1 comparison. Young and older CPS-weighted employment stocks are indexed to their own 2019 means; October 2025 remains missing. The families were selected after outcomes were observed because they generate the two largest absolute leave-family-out coefficient movements, so these paths are influence diagnostics rather than confirmatory evidence or a causal pandemic decomposition. The red dashed line marks January 2023.

Figure 2 shows the four direct-tail families, selected by descending nuisance-adjusted Q5 information rather than coefficient sign. Within each exposure tail, the young and older employment stocks are shown separately and indexed to their own 2019 means. The figure therefore exposes which age series drives a tail’s relative path without treating either age group as untreated.

Occupation leave-one-out calculations are interpreted in the same way. Fast-food and counter workers have high signed influence because low-exposure service employment recovered differently after the pandemic. Other occupations push in the opposite direction. Historical-contract diagnostics show that deleting the five largest absolute prior influences moves the coefficient toward zero, whereas deleting ten or twenty can make it

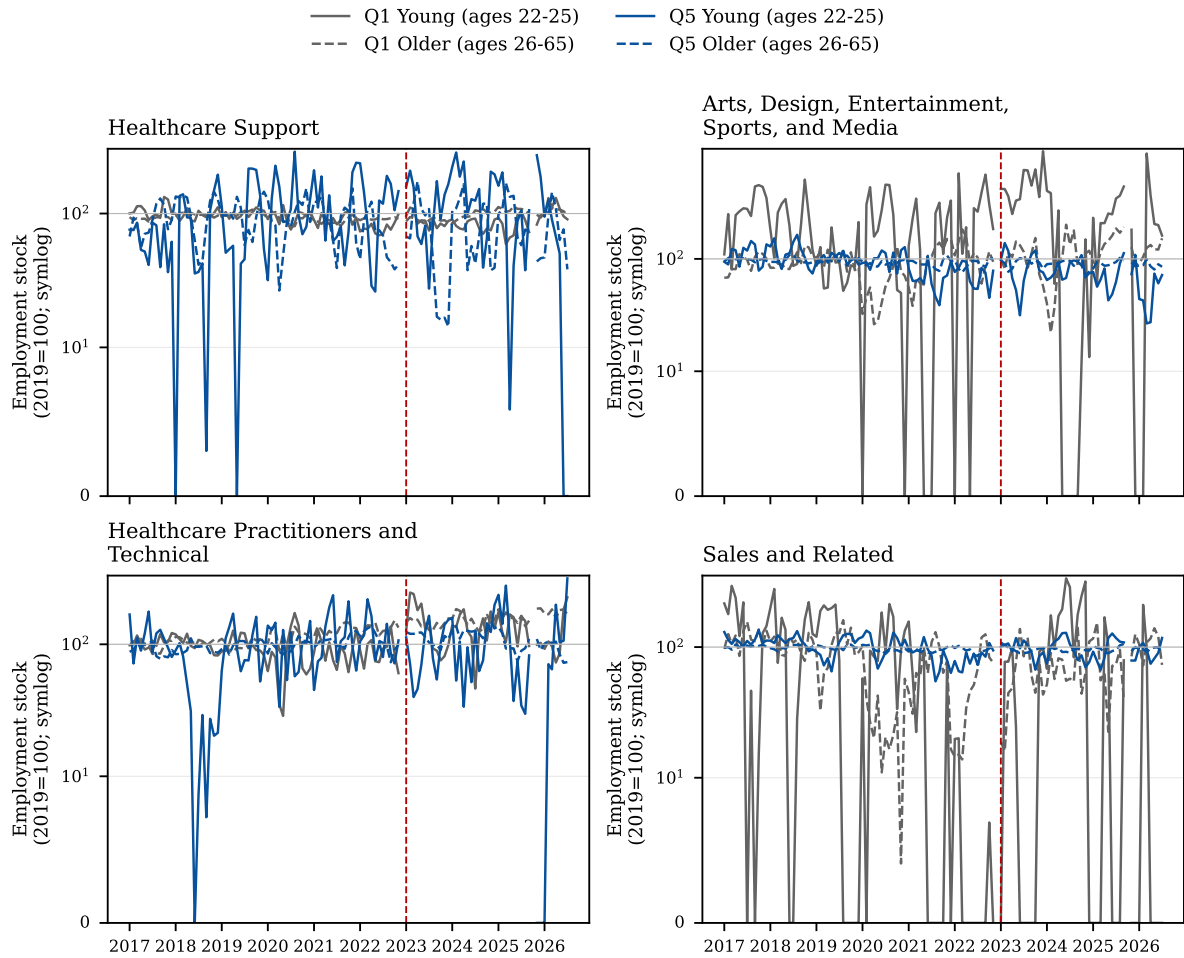


FIGURE 2. Young and older employment stocks in direct-tail families

Notes: Families are selected by direct-tail, SOC2-by-calendar-month Q5 nuisance-adjusted information, with ties broken by SOC2 code; selection does not use coefficient sign. Gray and blue distinguish Q1 and Q5; solid and dashed series distinguish ages 22–25 and 26–65. Each series equals 100 at its own 2019 mean. The symmetric-log vertical scale is linear from zero to 10 and logarithmic above 10 so that sparse-cell spikes do not obscure the other series; exact zero-stock months remain visible. The red dashed line marks January 2023. These are descriptive weighted-stock paths without design-based bands.

more negative. Symmetric influence trimming stays near the pooled estimate. Because every such deletion is outcome adaptive, none replaces the broad-family regression.

Economically defined service exclusions are cleaner diagnostics but still change the population. Removing Q1 food-service occupations attenuates the historical-contract absolute coefficient by less than one tenth; removing the broader SOC35/37/39 set of food, building-and-grounds, and personal-care service occupations does not attenuate it. Protective-service occupations (SOC33) are not part of that exclusion. These results reject a one-occupation explanation, not hypotheses about reopening, immigration, wages, or labor supply that the CPS design does not separately identify.

D. Occupational Characteristics and Heterogeneity

D.1. Inputs, timing, and common-support comparisons

The characteristic exercise uses one variable at a time before considering a parsimonious joint model. Every row first re-estimates the unaugmented exposure model on exactly the support available for that characteristic. It then adds the characteristic interacted with young status and the post indicator, reuses the same occupation multipliers, and forms the augmented-minus-baseline difference within each draw. Sample loss and coefficient movement are therefore distinct objects. The corresponding paired minimum detectable effect is the two-sided, five-percent normal-theory quantity $(1.96+0.84)SE(\Delta)$; it describes resolution and is never used as a second rejection rule.

The leading computer-use variable is O*NET item 4.A.3.b.1, “Interacting With Computers,” from release 24.3 (May 2020). Remotability is the Dingel–Neiman occupation classification. Wage and observed education composition are calculated from the CPS preperiod, while externally coded education requirements are kept conceptually separate. Routine and manual indices use preperiod O*NET inputs. All scales, source files, checksums, weighted correlations, and support counts are stored in the characteristic result directory.

The primary matched comparison uses all 455 occupations with computer-use ratings and crosses computer conditioning with family-month paths. The four Q5-by-post coefficients are -0.09575 without either adjustment, -0.19660 with computer use only, 0.03539 with family paths only, and -0.04665 with both. Relative to the pooled baseline, the corresponding paired movements are -0.10086 ($[-0.17723, -0.02448]$), 0.13114 ($[0.03517, 0.22710]$), and 0.04910 ($[-0.06418, 0.16237]$). The combined-minus-computer movement is 0.14995 ($[0.06039, 0.23951]$), while combined-minus-family is -0.08204 ($[-0.15461, -0.00947]$) under occupation clustering. These distinct conditional projec-

tions are not additive components.

The computer-use-by-post coefficient itself is 0.04602 per weighted standard deviation in the pooled computer-only model, with occupation interval [0.01171, 0.08032], and 0.04868 ([0.01053, 0.08684]) in the combined family-month model. The weighted standard deviation is 0.97290 O*NET points, and the beta-computer correlation is 0.7970. Adding computer use reduces Q5 information retention from 69.1 to 29.0 percent in the pooled model and from 42.5 to 30.0 percent in the family-month model. Thus coefficient movement and inflation occur in a thin residual comparison; the computer coefficient does not make computer use exogenous or purify the exposure coefficient as AI.

I also evaluated formal omitted-confounder sensitivity rather than inserting the grouped-binomial pseudo- R^2 into a linear formula. The coefficient-stability calculations of [Oster \(2019\)](#), the partial- R^2 framework of [Cinelli and Hazlett \(2020\)](#), and the endogenous-control selection model of [Diegert, Masten, and Poirier \(2022\)](#) are derived for linear regression targets with specific strength restrictions. The present target is one component of a vector-valued grouped-binomial fixed-effect projection with generated exposure groups; no verified mapping supplies those scalar inputs. A linear-probability companion would answer a different estimand. I therefore report no arbitrary breakdown value. This applicability ruling is not evidence that unobserved confounding is weak.

The separate common-characteristic block uses 347 occupations and holds its support and primary-universe assignments fixed. Relative to its -0.12372 baseline, computer use moves the coefficient by -0.12475 ($[-0.20934, -0.04016]$). The paired movements for remotability, wages, education requirements, routine intensity, and manual/physical content each include zero. Adding all six gives -0.23331 and a paired movement of -0.10958 ($[-0.22069, 0.00152]$). Family paths alone give -0.00348 ; adding all six characteristics to that model gives -0.10506 , with a paired movement of -0.10158 ($[-0.20057, -0.00258]$). The common-support block covers 76.78 percent of rebuilt treatment-construction stock and likewise does not identify a purified AI contrast.

D.2. Expanded support-specific and cumulative audit

The separate characteristic module in Tables [14](#) and [15](#) reports the support-specific comparisons and the uncontaminated portion of the cumulative sequence. Its one-at-a-time rows use each characteristic's maximal finite support, whereas its corrected joint rows use the literal 347-occupation common support. An audit found that an earlier version of this module had retained historical full-static-panel exposure groups and a 341-occupation historical intersection. The reported module now authenticates and uses

the canonical corrected-preperiod membership and Webb normalization throughout; its 468-occupation baseline exactly reproduces -0.1321 . The support-specific computer comparison retains 455 occupations and holds those rebuilt primary-universe groups fixed, while the focused exercise requires both computer-use and remotability data and rebuilds groups on its 408-occupation support. Their coefficients therefore differ by design, not by transcription. Rows at and after the superseded unbounded shortfall control are removed rather than mixed with the corrected shortfall population. Every addition and this corrective rerun are labeled post-outcome exploratory.

TABLE 14. Post-outcome exploratory characteristic-conditioning estimates

Conditioning block	Occupations	Baseline coefficient	Augmented coefficient [95% CI]	Paired change [95% CI]	Paired MDE ₈₀
<i>Panel A. One-at-a-time maximal-support comparisons</i>					
Computer use	455	-0.096	-0.197 [-0.316, -0.077]	-0.101 [-0.177, -0.024]	0.108
Remotability	408	-0.115	-0.122 [-0.222, -0.022]	-0.007 [-0.048, 0.033]	0.058
Preperiod wage	457	-0.144	-0.161 [-0.248, -0.075]	-0.017 [-0.044, 0.009]	0.038
Education requirement	404	-0.100	-0.105 [-0.203, -0.008]	-0.005 [-0.027, 0.016]	0.031
Routine-task intensity	449	-0.137	-0.129 [-0.221, -0.036]	0.008 [-0.011, 0.027]	0.027
Manual/physical content	455	-0.140	-0.142 [-0.289, 0.004]	-0.002 [-0.092, 0.089]	0.130
SOC2 by young by post	468	-0.132	-0.022 [-0.162, 0.119]	0.111 [0.008, 0.213]	0.146
<i>Panel B. One-at-a-time and joint comparisons on the 347-occupation common support</i>					
Computer use	347	-0.124	-0.248 [-0.391, -0.106]	-0.125 [-0.209, -0.040]	0.124
Remotability	347	-0.124	-0.137 [-0.239, -0.034]	-0.013 [-0.052, 0.026]	0.057
Preperiod wage	347	-0.124	-0.135 [-0.235, -0.036]	-0.012 [-0.029, 0.006]	0.025
Education requirement	347	-0.124	-0.134 [-0.235, -0.032]	-0.010 [-0.027, 0.006]	0.024
Routine-task intensity	347	-0.124	-0.112 [-0.220, -0.003]	0.012 [-0.024, 0.048]	0.051
Manual/physical content	347	-0.124	-0.122 [-0.255, 0.010]	0.001 [-0.079, 0.082]	0.117
All six characteristics	347	-0.124	-0.233 [-0.388, -0.078]	-0.110 [-0.221, 0.002]	0.164

Notes: All entries are post-outcome exploratory and were designed after the initial outcome pattern was known. Each baseline-augmented pair uses identical occupations, the canonical rebuilt corrected-preperiod beta groups and Webb normalization, and 9,999 common occupation-level Rademacher multipliers. Panel A uses each characteristic’s maximal finite subset; Panel B uses the current 347-occupation common support, which covers 76.78 percent of primary-support January 2017–November 2022 treatment-construction stock. For comparison, the unaugmented coefficient on all 468 occupations is -0.132 , while the unaugmented coefficient after imposing the 347-occupation common support is -0.124 ; that support-only movement is kept separate from every paired conditioning change. Rows from the superseded unbounded pandemic-shortfall construction are omitted; the corrected shortfall analysis uses a separate 363-occupation population and appears in Table 16. A confidence interval containing zero is a nondetection, not evidence of equivalence; conditioning does not identify a causal AI component.

D.3. Pandemic shortfall

The corrected exercise uses ages 22–65 throughout and admits only one-to-one occupation mappings with sufficient 2017–2019 histories. It retains 363 occupations, rather than the central 468. Linear trend predictions for total employment are bounded below at zero;

TABLE 15. Post-outcome exploratory support and target-information diagnostics

Conditioning block	Employment coverage (%)	Information retained (%)	Target VIF-like ratio	Effective occupation count	Top-five information share (%)
<i>Panel A. One-at-a-time maximal-support comparisons</i>					
Computer use	96.75	29.02	3.45	52.58	22.34
Remotability	88.58	52.66	1.90	29.25	30.65
Preperiod wage	97.93	59.45	1.68	37.86	26.32
Education requirement	84.54	63.60	1.57	27.48	31.24
Routine-task intensity	98.45	65.19	1.53	43.26	24.59
Manual/physical content	96.85	30.61	3.27	51.05	22.34
SOC2 by young by post	100.00	20.81	4.81	53.06	21.87
<i>Panel B. Cumulative comparisons on the 341-occupation common support</i>					
Computer use	76.77	29.33	3.41	48.88	21.98
+ remotability	76.77	28.46	3.51	44.97	24.72
+ wage, education, routine, and manual	76.77	21.08	4.74	58.00	20.28

Notes: All diagnostics are post-outcome exploratory. Employment coverage is the share of primary-support January 2017–November 2022 corrected treatment-construction stock retained. “Information retained” is nuisance-adjusted target information divided by fixed-effect-adjusted raw target information; its reciprocal is the reported target VIF-like ratio. These are design-information diagnostics, not an estimate of latent exposure reliability. The effective occupation count is the inverse concentration of occupation-level conditional target-information shares, and the final column is the share supplied by the five largest occupations. In the 341-occupation common-support sample, the 2017–2019-employment-weighted correlations of beta exposure with computer use, remotability, wage, education requirement, routine-task intensity, and manual/physical content are 0.807, 0.562, 0.429, 0.421, 0.197, and -0.748 , respectively. Rows based on the superseded unbounded pandemic-shortfall construction are omitted. Substantial information loss makes residual coefficients less precise and does not purify an AI treatment.

predicted young employment shares are bounded to $[0, 1]$. This is consequential: before bounding there are 174 negative total predictions and 1,258 out-of-range share predictions. The characteristics are the average bounded trend-prediction-minus-realized gaps from January 2020 through November 2022; December 2022 remains the transition month.

On this narrower common support, the pooled and family-month baseline coefficients are -0.16417 and -0.08708 . Total-stock conditioning moves them by 0.00040 and -0.00050 ; young-relative conditioning moves them by 0.00114 and 0.00613 . Table 16 reports linked-household sampling sensitivities that regenerate the shortfall inside each draw. All four conditioning-movement intervals include zero. Jointly regenerating exposure labels changes an average of 1.99 occupations per draw and leaves the young-relative family-month movement at 0.00613 with sampling-sensitivity standard error 0.00695 .

All 399 linked-household draws finish with zero failed joint refits. The largest binding bootstrap-of-bootstrap endpoint Monte Carlo standard error is 0.00979 log point, below the declared 0.01 threshold, so the predeclared expansion to 799 draws is not triggered.

The controls are generated from related CPS outcomes and may share sampling error and regression-to-the-mean with the dependent stocks. A deterministic household split gives a similar average but heterogeneous directions; it does not eliminate PSU-level common error. The bounded functional form and changed population also remain consequential. The exercise therefore does not establish that pandemic dynamics are absent, and the

linked-household intervals are not complete CPS design-based inference.

TABLE 16. Corrected bounded pandemic-shortfall diagnostics

Shortfall	Model	Baseline Q5–Q1	Conditioned Q5–Q1	Movement [sampling interval]
Total stock	Pooled	−0.16417	−0.16377	0.00040 [−0.00655, 0.00661]
Total stock	SOC2 × month	−0.08708	−0.08758	−0.00050 [−0.01377, 0.00970]
Young relative	Pooled	−0.16417	−0.16303	0.00114 [−0.00509, 0.00573]
Young relative	SOC2 × month	−0.08708	−0.08095	0.00613 [−0.00889, 0.01835]

Notes: The common population contains 363 one-to-one-mapped occupations with sufficient preperiod histories. Total-employment predictions are bounded below at zero and young-share predictions to $[0, 1]$. Movement is the conditioned minus baseline exposure coefficient. Brackets are basic intervals from 399 linked-household draws that regenerate the shortfall but hold exposure labels fixed. They are released-weight sampling sensitivities, not occupation-wild or complete CPS design-based intervals, and are not mechanically combined with the article’s shock-cluster variance. Joint label regeneration gives the same displayed point estimates and very similar uncertainty.

D.4. Industry, education, and exact age

The industry analysis is rebuilt at the occupation-by-broad-industry-by-age-by-month level rather than assigning each occupation a dominant industry. The risk set covers all 468 occupations and 3,127 preperiod-connected occupation–industry cells. Relative to the stratified baseline coefficient of -0.1376 , adding 13 broad-industry-specific young-by-post shifts gives -0.0987 . The paired movement is 0.0389 , with interval $[0.0082, 0.0697]$ and paired $MDE_{80} = 0.0444$. The conditioned coefficient remains below zero. These are statements about the microcell objective, not firm-level adoption.

Education-specific models are estimated on a common occupational support. The BA-or-higher estimate is -0.1380 and the non-BA estimate is -0.1226 ; their paired difference is -0.0154 , with interval $[-0.1557, 0.1250]$ and paired $MDE_{80} = 0.2014$. The comparison is too imprecise to distinguish moderate education differences. Exact-age estimates for ages 22, 23, 24, and 25 are reported as one jointly estimated exploratory collection with simultaneous intervals. No single age is selected after inspection.

D.5. Older comparison bands, age composition, and enrollment

Table 18 varies the older comparison group while holding the current exposure labels and occupational support fixed. The pooled coefficient remains negative across all five bands, but its magnitude is not constant. Against the 26–65 benchmark, paired movements for the 26–30, 31–40, and 41–50 comparisons have occupation-multiplier intervals

TABLE 17. Industry, education, and exact-age comparisons

Model	Occupations	Coefficient	Pointwise 95% interval	Simultaneous 95% interval
Occupation–industry–cell baseline	468	−0.138	[−0.228, −0.048]	—
Industry-conditioned	468	−0.099	[−0.188, −0.009]	—
BA or higher	429	−0.138	[−0.275, −0.001]	[−0.293, 0.017]
No BA	429	−0.123	[−0.231, −0.014]	[−0.243, −0.002]
Age 22	439	−0.093	[−0.212, 0.026]	[−0.239, 0.053]
Age 23	439	−0.140	[−0.252, −0.028]	[−0.280, −0.0003]
Age 24	439	−0.222	[−0.358, −0.086]	[−0.394, −0.050]
Age 25	439	−0.082	[−0.188, 0.024]	[−0.212, 0.047]

Notes: Both industry rows use occupation–broad-industry–age–month weighted-stock cells. The baseline includes common Q2–Q5 and Webb interactions; the conditioned row additionally includes 13 broad-industry-specific young-by-post shifts. Education and age collections use their own common occupational support and 9,999 common occupation multipliers. The four-age common-score equality test gives $\chi^2(3) = 8.21$, $p = .0419$, but every simultaneous age-minus-pooled interval includes zero; no individual age departure is promoted.

[−0.0412, 0.0644], [−0.0610, 0.0056], and [−0.0413, 0.0271]. The 51–65 movement is 0.0350 ([0.0036, 0.0663]). Under family-month conditioning, every coefficient interval and all four corresponding paired movement intervals include zero when shocks are assigned at the occupation level. Under 22-family shock assignment the ages 22–25 versus 26–30 paired movement instead has interval [−0.1712, −0.0041]; the two procedures answer different stochastic questions and are not combined, and neither is evidence of a causal age effect.

Fixing the older group’s exact-age composition at its preperiod shares gives a pooled coefficient of −0.1313 ([−0.2202, −0.0425]) and a family-month coefficient of −0.0218 ([−0.1597, 0.1161]). Relative to observed age composition, the paired movements are 0.0008 ([−0.0011, 0.0027]) and −0.0001 ([−0.0026, 0.0023]).

Enrollment restrictions change the numerator rather than assigning exposure to nonemployed people. With all ages 26–65 in the denominator, restricting ages 22–25 to nonenrolled workers changes the pooled coefficient from −0.1321 to −0.1239; the paired movement is 0.0082 ([−0.0125, 0.0289]). On the common 22–54 enrollment-question universe, restricting both age groups to nonenrolled workers changes the pooled coefficient from −0.1407 to −0.1354, a paired movement of 0.0053 ([−0.0144, 0.0250]). The analogous family-month movements are −0.0068 ([−0.0393, 0.0257]) and −0.0059 ([−0.0377, 0.0260]). Enrollment may itself respond to labor-market conditions, so these comparisons describe composition rather than a causal correction. Older workers throughout are comparison groups, not untreated controls.

TABLE 18. Older comparison bands and enrollment restrictions

Comparison	Pooled coefficient and 95% interval	Family-month coefficient and 95% interval
Ages 22–25 versus 26–65	–0.1321 [–0.2211, –0.0431]	–0.0217 [–0.1607, 0.1174]
Ages 22–25 versus 26–30	–0.1205 [–0.1881, –0.0530]	–0.1093 [–0.2328, 0.0141]
Ages 22–25 versus 31–40	–0.1598 [–0.2455, –0.0742]	–0.0247 [–0.1709, 0.1216]
Ages 22–25 versus 41–50	–0.1392 [–0.2344, –0.0440]	–0.0022 [–0.1429, 0.1385]
Ages 22–25 versus 51–65	–0.0971 [–0.2037, 0.0094]	–0.0021 [–0.1627, 0.1584]
Older exact-age composition fixed	–0.1313 [–0.2202, –0.0425]	–0.0218 [–0.1597, 0.1161]
Young numerator nonenrolled	–0.1239 [–0.2154, –0.0324]	–0.0285 [–0.1666, 0.1097]
Both groups nonenrolled, ages 22–54	–0.1354 [–0.2243, –0.0465]	–0.0356 [–0.1704, 0.0991]

Notes: Intervals use 9,999 occupation multipliers. The fixed-composition row holds the 26–65 exact-age shares at their preperiod values. The young-numerator row retains all employed ages 26–65 in the denominator. The last row restricts both groups to SCHLCOLL = 5 on the common ages 22–54 question universe. Older workers are comparison groups, not untreated controls; an interval containing zero indicates non-detection rather than equivalence.

D.6. Young-worker composition and national cohort supply

Table 19 keeps employment-conditioned composition separate from national cohort supply. The Q1–Q5 columns summarize employed ages 22–25 in occupations on the fixed 468-occupation support. The national profile uses every positive-weight CPS record ages 22–25 and assigns no occupation or exposure quintile. Across Q1–Q5, the pre-to-post enrolled-share changes are –0.10, –0.74, –2.00, –1.46, and –1.80 percentage points; BA-or-higher shares change by 1.79, 1.71, 2.69, 1.48, and 3.81 points. No employed-quintile mean-age change exceeds 0.017 year in absolute value, and the largest exact-age-share change is 1.62 points. Nationally, enrollment changes from 0.2336 to 0.2211, BA-or-higher from 0.2941 to 0.3121, employment from 0.7110 to 0.7247, and mean age by –0.0085 year; average monthly weighted population rises from 16.94 to 17.60 million. These are descriptive composition and selection patterns. Current enrollment may respond to labor-market conditions, and the profiles do not identify a cohort correction or assign exposure to nonworkers.

E. Inference and Finite-Sample Audit

E.1. Stochastic targets

The principal wild-score interval treats occupation as the shock cluster and conditions on the constructed stocks, exposure labels, occupational bridge, and retained support. It accommodates arbitrary serial dependence within an occupation but not unrestricted dependence across occupations. A 22-family Webb-weight exercise instead allows common

TABLE 19. Education, enrollment, and age composition of ages 22–25*Panel A: Annual enrollment and BA-or-higher shares*

Year	Q1	Q2	Q3	Q4	Q5	National population	National population (millions/month)
2017	.135/.085	.201/.239	.191/.397	.142/.473	.171/.477	.244/.275	17.376
2018	.113/.089	.207/.232	.188/.439	.149/.470	.171/.484	.239/.281	17.295
2019	.121/.080	.208/.238	.168/.449	.151/.489	.162/.518	.236/.293	16.865
2020	.121/.098	.190/.247	.173/.463	.125/.517	.158/.562	.232/.314	16.848
2021	.118/.101	.177/.245	.167/.444	.149/.474	.164/.555	.223/.301	16.541
2022	.120/.088	.186/.253	.159/.459	.130/.489	.152/.538	.225/.302	16.673
2023	.118/.110	.181/.245	.154/.470	.123/.480	.152/.534	.219/.306	17.659
2024	.121/.109	.195/.249	.147/.469	.129/.517	.136/.559	.221/.315	17.369
2025	.121/.113	.189/.269	.157/.463	.128/.493	.151/.580	.222/.316	17.638
2026	.124/.096	.186/.283	.164/.470	.127/.516	.140/.575	.222/.313	17.815

Panel B: Preperiod/postperiod age composition

Group	Mean age	YEAR–age proxy	Age 22	Age 23	Age 24	Age 25
Q1	23.482/23.486	1995.9/2000.8	.257/.251	.251/.250	.245/.261	.247/.238
Q2	23.481/23.469	1995.9/2000.9	.256/.253	.253/.257	.244/.258	.246/.232
Q3	23.642/23.638	1995.8/2000.7	.198/.200	.248/.241	.267/.279	.286/.280
Q4	23.690/23.681	1995.8/2000.6	.190/.184	.235/.245	.270/.276	.305/.295
Q5	23.657/23.674	1995.8/2000.6	.194/.192	.241/.236	.280/.279	.286/.293
National	23.514/23.506	1995.9/2000.8	.247/.246	.248/.249	.250/.258	.255/.247

Notes: Q1–Q5 condition on employment and the fixed 468-occupation support. “National population” uses every positive-WTFINL CPS record ages 22–25 and assigns neither an occupation nor an exposure quintile to nonworkers. Entries in Panel A before the final column are enrollment share/BA-or-higher share; entries in Panel B are preperiod/postperiod. The preperiod is January 2017–November 2022 and the post-period January 2023–July 2026; December 2022 is omitted, October 2025 is absent, and 2026 is partial. Annual national population is the person-weight sum divided by observed months. YEAR minus age is an approximate birth-year proxy, not an age–period–cohort coefficient. Complete annual age/cohort cells remain in the machine-readable composition files.

shocks within broad SOC families. A separate household/sample-unit multiplier rebuilds cell stocks from microdata, preserves all observed records of the sampled unit across months, and assigns the same multiplier to every fractional descendant of a routed source record. That exercise describes repeated-sample sensitivity conditional on released CPS final weights; the public Basic Monthly extracts do not supply the design variables needed to call it complete CPS design-based inference.

These variances are not mechanically added. Occupation-shock and household-resampling procedures can contain overlapping components, so summing them without a derived orthogonal decomposition could double count uncertainty. They are reported as separate sensitivity targets.

E.2. Score and bootstrap implementation

The conditional likelihood retains one-sided zero cells and drops only occupation-month cells with zero total stock. Scores use the fitted grouped-binomial objective, and nuisance-adjusted target scores use the Schur-complement projection implied by the fitted information matrix. The same 9,999 Rademacher occupation multipliers are reused for paired comparisons. Each draw first forms the two refitted or score-updated targets and then their difference; intervals for movement therefore retain covariance. The finite-cluster correction, random seed, multiplier matrix hash, convergence checks, and score-conservation checks are written to the execution receipts.

For the reconstructed pooled comparison, the analytic standard error is 0.04517 and the 9,999-draw 95-percent interval is $[-0.2206, -0.0437]$. After SOC2-by-calendar-month conditioning, the coefficient is -0.0217 , with standard error 0.07132 and interval $[-0.1607, 0.1173]$. The paired movement is 0.11043, with standard error 0.05189 and interval $[0.0107, 0.2102]$. The corresponding two-sided normal-theory paired MDE_{80} is 0.14539. The detected movement remains smaller than this planning quantity; that is not a contradiction because an MDE is a power summary, not an acceptance threshold for an observed draw.

Table 20 consolidates the principal precision diagnostics. It states the scale and whether the quantity is a coefficient or a paired difference; the same numerical formula is not reinterpreted as an equivalence margin.

E.3. Numerical existence and separation

All eleven declared grouped-binomial specifications pass the treatment-rank, recession-direction, binding trust-path versus independent zero-start Newton/IRLS agreement,

TABLE 20. Consolidated normal-theory precision diagnostics

Analysis	Quantity	Scale	MDE ₈₀
Core pooled	Q5–Q1 coefficient	Log young-to-older stock ratio	0.1266
Family-month	Q5–Q1 coefficient	Log young-to-older stock ratio	0.1998
Family-month vs. pooled	Paired coefficient difference	Log young-to-older stock ratio	0.1454
Direct tail within family	Q5–Q1 coefficient	Log young-to-older stock ratio	0.4576
Continuous within family	Exposure slope	One within-family exposure SD	0.0313
Computer-use conditioning	Paired coefficient difference	Log young-to-older stock ratio	0.1131
Remotability conditioning	Paired coefficient difference	Log young-to-older stock ratio	0.0510
Industry conditioning	Paired coefficient difference	Log young-to-older stock ratio	0.0444
BA vs. non-BA	Paired coefficient difference	Log young-to-older stock ratio	0.2014
Architecture comparisons	Paired coefficient differences	Log young-to-older stock ratio	0.0609–0.1689
Adjacent exit	Q5–Q1 coefficient	Log conditional transition-rate ratio	0.254
Adjacent occupation change	Q5–Q1 coefficient	Log conditional transition-rate ratio	0.174
Adjacent entry destination	Q5–Q1 coefficient	Log conditional allocation ratio	0.260
Twelve-month exit	Q5–Q1 coefficient	Log conditional transition-rate ratio	0.336
Twelve-month occupation change	Q5–Q1 coefficient	Log conditional transition-rate ratio	0.115
Twelve-month entry destination	Q5–Q1 coefficient	Log conditional allocation ratio	0.282

Notes: $\text{MDE}_{80} = (1.96 + 0.84)\widehat{SE}$ is the two-sided five-percent normal-theory minimum detectable quantity at 80-percent power. It summarizes resolution on the displayed scale; it is not a wild-score critical value, equivalence margin, or acceptance threshold. Architecture entries report the observed range across seven beta-minus-alternative comparisons. Flow rows use their own risk sets and should not be compared as if they estimated one common probability.

fitted-mean, normalized-gradient, raw and diagonally scaled fitted-Hessian, and two-sided target-profile checks at the unchanged A1 thresholds. The pooled, family-post, family-month, dynamic, and quintile-by-month seasonality specifications require no boundary-row profiling. Each post-2020 specification profiles one boundary nuisance group spanning 77 positive-total data rows, and each occupation-by-month seasonality specification profiles 110 boundary nuisance groups spanning 770 rows; these rows are profiled on the extended-likelihood face rather than deleted. In every model the recession audit finds no direction capable of moving a reported target or treatment coefficient, and the focal target is finite on the resolved extended-likelihood face. The retained L-BFGS-B path is diagnostic rather than a passing solver: it fails the standardized-score threshold in all eleven fits but reveals no better independently stationary contradictory target.

E.4. Broad-family and household sensitivity

The few-family exercise uses 22 broad-family weights and reports baseline, conditioned, and paired targets from common draws. Its role is sensitivity to cross-occupation dependence, not a declaration that SOC2 is the uniquely correct cluster. The household/sample-unit exercise performs 399 positive-weight full refits. With labels fixed, its basic interval is $[-0.1865, -0.0794]$ for the pooled coefficient, $[-0.1212, 0.0587]$ for the family-conditioned coefficient, and $[0.0362, 0.1753]$ for their movement. Regenerating preperiod exposure labels within the same draws gives $[-0.1864, -0.0762]$, $[-0.1236, 0.0665]$, and $[0.0355, 0.1771]$. A median of two occupations changes quintile per regenerated-label draw. Webb re-standardization by itself is an affine transformation absorbed by the specified fixed effects, so this difference is a quintile-membership sensitivity rather than a separate normalization channel.

All 399 household refits succeed. The authoritative stopping calculation puts the largest bootstrap-of-bootstrap interval-endpoint Monte Carlo standard error at 0.00820 log point, below the declared 0.01 threshold, so the predeclared expansion to 799 draws is not triggered.

The source does not expose official Basic Monthly replicate weights or all confidential design identifiers. Household identifiers are therefore a dependence-preserving public-data approximation, not an ASEC replication-weight procedure. Link-specific flow analyses additionally report person- and household-cluster sensitivities using their own risk sets.

E.5. Calendar-time HAC and stress validation

The time-HAC audit operates on nuisance-adjusted score sums and implements occupation clustering plus an aggregate elapsed-calendar HAC term minus within-occupation elapsed-calendar HAC terms. Calendar lags are measured in months, so the missing October 2025 observation does not make September and November adjacent. Under rebuilt exposure labels, lag-16 standard errors are 0.04449 for the pooled target, 0.05522 for the family-conditioned target, and 0.04501 for their paired movement. None of the 15 rebuilt five-parameter covariance matrices across the three targets and five lags is indefinite, and no projection or eigenvalue clipping is applied. Indefinite positive-lag matrices from the superseded historical-treatment/SOC2-post audit are retained only as labeled implementation history.

The completed finite-sample audit uses 11 declared designs, 799–1,599 retained outer replications per design, 9,999 common multiplier draws per joint refit, and independently computed pseudo-targets. Absolute coefficient bias is at most 0.00102 log point in the empirical observed-effect design. Interval performance is target-specific: occupation-Rademacher coverage is 0.865 for the pooled projection, 0.921 for the family-month projection, and 0.889 for their paired movement. Family-Rademacher coverage is respectively 0.990, 0.895, and 0.992. Rademacher and Webb weights give nearly identical coverage at a fixed cluster level, so changing the multiplier law does not repair the cluster-level mismatch.

The earlier 26.7-percent and 11.3-percent rejection figures are withdrawn rather than reinterpreted. In the clean family-variance-zero design, the family-month pseudo-target is zero to numerical precision; its five-percent rejection rates are 6.1 percent under occupation Rademacher weights, 8.9 percent under family Rademacher weights, and 5.6 percent for the cross-fit full-refit oracle. In the empirically calibrated structural-null design, removing the structural exposure coefficient still leaves a pooled pseudo-target of -0.11035 and a paired pseudo-target of 0.11035 because the data-generating process retains the observed fitted family-by-month surface. The near-nominal oracle then rejects zero for the pooled projection 96.5 percent of the time. This is a restatement of the broad-family decomposition built into the calibration, not independent evidence that composition caused the empirical gap. The simulations therefore limit both the inferential and structural interpretation: none of the feasible cluster procedures is adequate for every target and significance of the pooled projection does not by itself test a structural exposure effect.

Tables 21– 23 make the simulation inventory and its Monte Carlo uncertainty explicit. The first compares every declared procedure on a common empirically calibrated design;

the next two report estimation and interval performance for all eleven core, adverse, and factor-ablation designs while holding the occupation-Rademacher procedure fixed. This separation prevents a change in the DGP from being mistaken for a change in the cluster or multiplier rule.

TABLE 21. Finite-sample performance in the empirical observed-effect design

Target	Procedure	θ^*	Bias	Empirical SD	Mean SE	Coverage (MCSE)	Reject 0 (MCSE)	Mean length	Failed/N
Pooled	Occupation Rademacher	-0.13198	+0.00065	0.02790	0.02173	0.865 (0.012)	1.000 (0.000)	0.08439	0/799
Pooled	Occupation Webb	-0.13198	+0.00065	0.02790	0.02173	0.869 (0.012)	1.000 (0.000)	0.08495	0/799
Pooled	Family Rademacher	-0.13198	+0.00065	0.02790	0.03845	0.990 (0.004)	0.977 (0.005)	0.14806	0/799
Pooled	Family Webb	-0.13198	+0.00065	0.02790	0.03845	0.990 (0.004)	0.977 (0.005)	0.14814	0/799
Pooled	Cross-fit oracle	-0.13198	+0.00065	0.02790	0.02173	0.945 (0.008)	0.999 (0.001)	0.10663	0/799
Family-month	Occupation Rademacher	-0.02165	+0.00101	0.02352	0.02175	0.921 (0.010)	0.169 (0.013)	0.08488	0/799
Family-month	Occupation Webb	-0.02165	+0.00101	0.02352	0.02175	0.921 (0.010)	0.169 (0.013)	0.08487	0/799
Family-month	Family Rademacher	-0.02165	+0.00101	0.02352	0.02114	0.895 (0.011)	0.190 (0.014)	0.08102	0/799
Family-month	Family Webb	-0.02165	+0.00101	0.02352	0.02114	0.891 (0.011)	0.189 (0.014)	0.08134	0/799
Family-month	Cross-fit oracle	-0.02165	+0.00101	0.02352	0.02175	0.949 (0.008)	0.121 (0.012)	0.09519	0/799
Family minus pooled	Occupation Rademacher	0.11034	+0.00036	0.03092	0.02585	0.889 (0.011)	0.979 (0.005)	0.10082	0/799
Family minus pooled	Occupation Webb	0.11034	+0.00036	0.03092	0.02585	0.892 (0.011)	0.977 (0.005)	0.10135	0/799
Family minus pooled	Family Rademacher	0.11034	+0.00036	0.03092	0.04109	0.992 (0.003)	0.836 (0.013)	0.15848	0/799
Family minus pooled	Family Webb	0.11034	+0.00036	0.03092	0.04109	0.992 (0.003)	0.834 (0.013)	0.15865	0/799
Family minus pooled	Cross-fit oracle	0.11034	+0.00036	0.03092	0.02585	0.945 (0.008)	0.944 (0.008)	0.11853	0/799

Notes: Movement is family-month minus pooled. The table reports all five procedures in the empirically calibrated design with the observed structural input. θ^* is the independently computed pseudo-true projection target. Parentheses give binomial Monte Carlo standard errors. Reject 0 is not an empirical-size statistic here because none of the three pseudo-targets is zero. The largest relative Monte Carlo error of an empirical SD in this panel is 0.02675. The cross-fit full-refit oracle learns critical values on one deterministic half of the replications and evaluates them on the other; it is a benchmark, not an implementable article interval. Every joint refit succeeded, and each multiplier procedure used 9,999 common draws.

F. Dynamics, Endpoints, and Trend Sensitivity

F.1. Quarterly system

The corrected dynamic model interacts every nonreference exposure quintile with calendar quarter. The omitted period is 2022Q4, which contains October and November because December 2022 is the transition month and is excluded. The resulting Q5–Q1 path has 23 preperiod and 15 postperiod coefficients. The pooled and SOC2-by-calendar-month versions are estimated on identical reconstructed support, and the archive stores the complete Q2–Q5 coefficient vector, full covariance matrix, influence matrix, and simultaneous critical values.

Relative to omitted 2022Q4, the joint null that all 23 Q5 preperiod coefficients equal zero is rejected in the pooled model ($p = .0171$) and the family-conditioned model ($p = .0053$). Table 24 also reports the reference-invariant null that the coefficients are equal to

TABLE 22. Declared simulation designs: occupation-Rademacher estimation performance

Design	Structural θ	N	Pseudo-truth	Bias	Empirical SD	Mean SE	Failed/ N	SD MC error
Empirical, null	0	1,199	-0.1103/0.0000/0.1104	+0.0003/ + 0.0003/ - 0.0000	0.0283/0.0236/0.0319	0.0218/0.0216/0.0257	0/1,199	0.0213
Empirical, local	-0.0500	799	-0.1603/ - 0.0499/0.1103	+0.0004/ - 0.0001/ - 0.0005	0.0273/0.0236/0.0317	0.0217/0.0218/0.0259	0/799	0.0261
Empirical, observed	-0.0217	799	-0.1320/ - 0.0216/0.1103	+0.0006/ + 0.0010/ + 0.0004	0.0279/0.0235/0.0309	0.0217/0.0217/0.0259	0/799	0.0267
Prior adverse, null	0	1,599	-0.0002/0.0001/0.0003	+0.0004/ - 0.0005/ - 0.0009	0.0327/0.0238/0.0355	0.0174/0.0216/0.0222	0/1,599	0.0184
Prior adverse, local	-0.0500	1,599	-0.0501/ - 0.0498/0.0003	+0.0001/ - 0.0004/ - 0.0004	0.0332/0.0238/0.0360	0.0174/0.0216/0.0223	0/1,599	0.0183
Prior adverse, observed	-0.1321	1,599	-0.1322/ - 0.1319/0.0003	+0.0005/ + 0.0002/ - 0.0003	0.0330/0.0237/0.0354	0.0176/0.0217/0.0223	0/1,599	0.0186
Equalized sparsity, null	0	1,199	-0.1103/0.0000/0.1104	+0.0002/ + 0.0002/ + 0.0000	0.0276/0.0189/0.0300	0.0207/0.0155/0.0227	0/1,199	0.0212
No family variance, null	0	799	-0.1105/ - 0.0000/0.1105	+0.0007/ - 0.0002/ - 0.0009	0.0123/0.0235/0.0205	0.0196/0.0217/0.0241	0/799	0.0276
No serial persistence, null	0	799	-0.1103/0.0000/0.1104	-0.0002/ - 0.0008/ - 0.0007	0.0190/0.0242/0.0245	0.0203/0.0218/0.0246	0/799	0.0267
Equalized influence, null	0	799	-0.0946/ - 0.0000/0.0946	+0.0014/ + 0.0028/ + 0.0014	0.0353/0.0486/0.0423	0.0293/0.0447/0.0364	0/799	0.0266
Occupation AR(1), null	0	799	-0.1103/0.0000/0.1104	-0.0007/ - 0.0004/ + 0.0004	0.0169/0.0307/0.0259	0.0224/0.0271/0.0279	0/799	0.0253

Notes: Within every slash-delimited cell the order is pooled, family-month, and movement (family-month minus pooled). This table holds the article procedure fixed at occupation-cluster Rademacher weights to isolate changes in the DGP; Table 21 compares all five procedures in the empirical observed-effect design. The empirical designs use actual support, totals, Kish counts, fitted nuisance indices, and a calibrated stationary family-month Gaussian AR(1) shock. The prior-adverse rows reproduce the archived sparse-cell and family-sign-shock design; all 195, 196, and 196 archived successful null, local, and observed draws match before extension, with maximum coefficient difference below 1.8×10^{-16} . The remaining rows separately equalize variance-preserving effective counts, remove family variance, remove serial persistence while holding marginal variance fixed, equalize information within family, or impose occupation-level AR(1) shocks. SD MC error is the largest relative Monte Carlo error of the three empirical SDs. Every joint refit succeeded.

TABLE 23. Declared simulation designs: occupation-Rademacher interval performance

Design	Coverage	Reject 0	Mean interval length	Binomial MCSE max
Empirical, null	0.852/0.916/0.877	0.992/0.084/0.975	0.0845/0.0842/0.1002	0.0103
Empirical, local	0.872/0.926/0.881	1.000/0.627/0.970	0.0843/0.0850/0.1008	0.0171
Empirical, observed	0.865/0.921/0.889	1.000/0.169/0.979	0.0844/0.0849/0.1008	0.0133
Prior adverse, null	0.686/0.917/0.775	0.313/0.083/0.227	0.0677/0.0841/0.0868	0.0116
Prior adverse, local	0.680/0.926/0.757	0.675/0.629/0.244	0.0678/0.0844/0.0870	0.0121
Prior adverse, observed	0.674/0.921/0.764	1.000/1.000/0.235	0.0684/0.0848/0.0872	0.0117
Equalized sparsity, null	0.835/0.883/0.859	0.993/0.117/0.982	0.0799/0.0604/0.0881	0.0107
No family variance, null	0.995/0.939/0.977	1.000/0.061 [†] /0.995	0.0761/0.0847/0.0941	0.0085
No serial persistence, null	0.959/0.934/0.954	1.000/0.066/0.994	0.0790/0.0851/0.0961	0.0088
Equalized influence, null	0.890/0.929/0.902	0.840/0.071/0.717	0.1147/0.1733/0.1416	0.0159
Occupation AR(1), null	0.990/0.909/0.960	1.000/0.091/0.977	0.0868/0.1054/0.1086	0.0102

Notes: Slash order is pooled, family-month, and movement (family-month minus pooled). Binomial MCSE max is the largest of the coverage and zero-rejection Monte Carlo standard errors shown in the complete result matrix. Only the daggered rejection rate is empirical size: its independently computed family-month pseudo-target is -1.31×10^{-16} . All other zero-rejection entries are rejection frequencies for nonzero pseudo-targets, including values that round to 0.0000 in Table 22, and they do not enter the size stopping rule. Thus a displayed maximum may exceed the 0.0125 stopping threshold when it belongs only to a nonzero-target rejection rate. All designs satisfied the declared accuracy rule before the 1,999-replication cap. Each multiplier procedure used 9,999 common draws. The complete 165-row five-procedure matrix is retained in ALL_SIMULATION_SUMMARY.csv.

one another within each fixed window. That null rejects over the full period for the pooled model ($p = .0121$) but not for the family-conditioned model ($p = .0641$); it does not reject at five percent in 2017Q1–2019Q4, 2021Q1–2022Q3, or the full window excluding 2020Q2–2020Q4 for either structure. Thus the full pooled rejection no longer clears five percent when 2020Q2–2020Q4 are excluded, whereas the family reference-relative rejection persists even though equality within each fixed window is not rejected. In leave-one-quarter diagnostics, 2019Q2 produces the largest reduction in the pooled reference-relative Wald statistic and 2018Q4 the largest reduction in the conditioned statistic; these nonadditive deletions do not identify unique causal drivers.

TABLE 24. Preperiod joint tests across fixed calendar windows

Structure	Window	Equal to 2022Q4	Equal within window
Pooled	2017Q1–2022Q3	0.0171	0.0121
Pooled	2017Q1–2019Q4	0.1365	0.0925
Pooled	2021Q1–2022Q3	0.4285	0.3281
Pooled	Full, excluding 2020Q2–Q4	0.0772	0.0676
Family by month	2017Q1–2022Q3	0.0053	0.0641
Family by month	2017Q1–2019Q4	0.0126	0.0744
Family by month	2021Q1–2022Q3	0.0116	0.3264
Family by month	Full, excluding 2020Q2–Q4	0.0079	0.1048

Notes: Entries are common-multiplier p -values. The third column tests every selected preperiod coefficient against the omitted 2022Q4 coefficient, which represents October–November 2022. The fourth tests equality among coefficients inside the selected window and is invariant to the omitted reference. The two nulls therefore answer different questions. Windows were declared before this result audit and are not selected for favorable p -values.

As a precision diagnostic, an OLS linear functional through all preperiod coefficients is -0.0078 log point per year in the pooled model and 0.0041 after family conditioning. Its occupation-cluster standard errors are 0.0119 and 0.0292 , implying normal-theory MDE_{80} values of 0.0332 and 0.0819 log point per year. An insignificant linear slope cannot rule out economically relevant smooth drift and is not a substitute for the joint tests. Simultaneous bands exclude zero only in late pooled quarters and never in the family-conditioned path. These results do not establish which preperiod force produces the differences and do not validate parallel evolution for a causal interpretation of the January 2023 split.

Table 25 distinguishes three targets. The postperiod functional P averages the 15 post-quarter coefficients with predeclared calendar weights relative to omitted 2022Q4. Its pooled value is -0.1199 , with interval $[-0.2637, 0.0239]$; its family-conditioned value

is -0.2074 , with interval $[-0.4252, 0.0103]$. The reference-invariant dynamic functional D subtracts the corresponding weighted preperiod mean. It is -0.131372 pooled and -0.021689 after family conditioning, compared with static coefficients S of -0.132109 and -0.021675 . Covariance-preserving tests give $p = .454$ and $p = .995$ for $D - S$. Hence the static and dynamic post-minus-pre targets are numerically reconciled. Within the family model, $P - S = -0.1858$ has $p = .074$, so the design does not detect that difference at five percent. Across conditioning structures, however, the change in $P - S$ is -0.1980 ($p = .021$), whereas the change in $D - S$ is -0.0008 ($p = .693$). Thus the short-reference functional responds to conditioning differently from the reference-invariant comparison. The $D - S$ result is a nondetection, not an equivalence test, and D remains a linear functional rather than the nonlinear static grouped-binomial coefficient.

TABLE 25. Static and dynamic target reconciliation

Specification	Static S	Post/reference P	Post/pre D	$D - S$	$p(D - S = 0)$
Pooled	-0.132109	-0.119889	-0.131372	0.000737	0.454
Family by calendar month	-0.021675	-0.207434	-0.021689	-0.000014	0.995
Family minus pooled	0.110434	-0.087545	0.109683	-0.000752	0.693

Notes: S is the static grouped-binomial Q5-by-post coefficient. P is the observed-post-month-weighted average of quarterly Q5 coefficients relative to the two observed months in omitted 2022Q4. D subtracts the corresponding observed-pre-month-weighted average and is invariant to the omitted event-time reference. The last two columns use the stored joint covariance, so they preserve the covariance between the static and dynamic fits. The tests establish nondetection of a difference, not economic equivalence.

F.2. Official HonestDiD implementation

The trend-sensitivity program passes the joint event-time vector and covariance matrix to the official HonestDiD implementation. The environment receipt fixes HonestDiD 0.2.8 and its source commit, and records the compatible official CVXR and solver versions. The program does not feed a single static coefficient, D , or a standard error into an event-study procedure; its target is P . It reports both the complete 2017Q1–2022Q3 calibration and the prespecified consecutive 2021Q1–2022Q3 post-reopening calibration. The latter does not delete an internal pandemic quarter or compress a calendar gap. For the family-conditioned vector its $M = 0$ smoothness interval excludes zero, but the interval includes zero at $M = 0.005$; the pooled intervals include zero even at $M = 0$. The conventional event-functional interval already includes zero in both structures. The correct positive zero-exclusion breakdown is therefore undefined: there is no positive robustness threshold to lose. Earlier installation and API failures remain in the execution log; the successful

official run is not replaced with a home-built approximation.

TABLE 26. Official HonestDiD sensitivity intervals

Structure	Calibration	Restriction	Lower	Upper
Pooled	2017Q1–2022Q3	Conventional	−0.264	0.024
		Smoothness, $M = 0$	−0.140	0.016
		Smoothness, $M = 0.005$	−0.482	0.516
		Relative magnitude, $\bar{M} = 0$	−0.260	0.019
		Relative magnitude, $\bar{M} = 0.5$	−1.023	0.789
Pooled	2021Q1–2022Q3	Conventional	−0.264	0.024
		Smoothness, $M = 0$	−0.230	0.204
		Smoothness, $M = 0.005$	−0.540	0.495
		Relative magnitude, $\bar{M} = 0$	−0.260	0.019
		Relative magnitude, $\bar{M} = 0.5$	−0.985	0.742
SOC2 monthly paths	2017Q1–2022Q3	Conventional	−0.425	0.010
		Smoothness, $M = 0$	−0.351	0.007
		Smoothness, $M = 0.005$	−0.869	0.352
		Relative magnitude, $\bar{M} = 0$	−0.420	0.007
		Relative magnitude, $\bar{M} = 0.5$	−2.222	1.884
SOC2 monthly paths	2021Q1–2022Q3	Conventional	−0.425	0.010
		Smoothness, $M = 0$	−0.772	−0.007
		Smoothness, $M = 0.005$	−1.100	0.207
		Relative magnitude, $\bar{M} = 0$	−0.420	0.007
		Relative magnitude, $\bar{M} = 0.5$	−2.222	1.884

Notes: The target is the observed-month-weighted average of the 15 post-quarter Q5–Q1 coefficients relative to 2022Q4. Intervals use the full occupation-cluster covariance and official HonestDiD 0.2.8. Smoothness M is in log points per quarter squared; $\bar{M}$ bounds post changes relative to the largest signed preperiod change. Both the full 2017Q1–2022Q3 calibration and the consecutive post-reopening 2021Q1–2022Q3 calibration are fixed and reported. The recent-window family-path interval excludes zero only at the exact-linear-bias restriction $M = 0$ and includes zero at $M = 0.005$. Because each conventional interval includes zero, positive zero-exclusion robustness is undefined, not an omitted threshold.

F.3. Onset, endpoint, seasonality, and survey dates

Onset sensitivity reports every monthly split from November 2022 through June 2023. January 2023 is retained because December 2022 was designated as the transition month after the November 30, 2022 public release of ChatGPT; robustness across nearby dates does not identify that event as the cause. The endpoint grid includes December 2024, September and December 2025, and July 2026 wherever the calendar permits. October 2025 is

never imputed. Relative to the full window, the pooled movements through December 2024, September 2025, and December 2025 are 0.0211, 0.0225, and 0.0212 log point. Their magnitudes are essentially the same even though their interval endpoints differ. The full available window remains the canonical target, while December 2024 is reported prominently as the pre-2025-production-break benchmark; the latter is not promoted after outcomes are observed because it retains a negative detected coefficient. The analogous conditioned movements remain near zero.

TABLE 27. Post-outcome exploratory endpoint sensitivity

Endpoint	Months	Coefficient [95% CI]	Endpoint – full [paired 95% CI]	Paired MDE ₈₀
<i>Pooled</i>				
Through December 2024	95	-0.1110 [-0.2008, -0.0213]	0.0211 [-0.0073, 0.0495]	0.0407
Through September 2025	104	-0.1096 [-0.1978, -0.0214]	0.0225 [0.0053, 0.0397]	0.0245
Through December 2025 (actual gap)	106	-0.1109 [-0.1979, -0.0239]	0.0212 [0.0057, 0.0367]	0.0221
Through July 2026, excluding September and November 2025	111	-0.1337 [-0.2236, -0.0438]	-0.0016 [-0.0075, 0.0043]	0.0084
Through July 2026 (reference)	113	-0.1321 [-0.2206, -0.0437]	Reference	—
<i>SOC2 by calendar month</i>				
Through December 2024	95	-0.0120 [-0.1657, 0.1417]	0.0097 [-0.0526, 0.0719]	0.0897
Through September 2025	104	-0.0219 [-0.1616, 0.1178]	-0.0002 [-0.0345, 0.0341]	0.0491
Through December 2025 (actual gap)	106	-0.0170 [-0.1555, 0.1215]	0.0047 [-0.0248, 0.0342]	0.0422
Through July 2026, excluding September and November 2025	111	-0.0252 [-0.1660, 0.1155]	-0.0035 [-0.0131, 0.0060]	0.0137
Through July 2026 (reference)	113	-0.0217 [-0.1607, 0.1173]	Reference	—

Notes: All rows use the rebuilt corrected-preperiod treatment and the same occupation support. The December 2024 row is the requested pre-2025 endpoint; the December 2025 row retains the actual missing October survey month. The shutdown-month row excludes September and November 2025 and does not interpolate October. Nonreference endpoint intervals and paired differences use the common endpoint-grid draws. To avoid printing a second seed-dependent interval for an identical target, each full-window reference uses its declared canonical single-target interval. The MDE is the two-sided five-percent, 80-percent normal-theory precision diagnostic, not an economic-equivalence threshold.

Exposure-group-by-month-of-year interactions and the feasible occupation-specific month-of-year specification both move the target little: the largest absolute movement from the corresponding full-window coefficient is 0.0011. A coding-stable post-2020 specification gives -0.1181 pooled and -0.0304 with family-month effects; paired movements from the full-window estimates are 0.0140 ([-0.0288, 0.0569]) and -0.0087 ([-0.1053, 0.0878]). These replacement fits pass the same numerical certification as the baseline. The earlier failed attempts and diagnostics remain retained in the failure registry. Excluding September and November 2025 around the collection gap does not detectably change the full-window estimate. These rows address sensitivity to calendar construction and survey production, not counterfactual reweighting for the January 2025 or January 2026 population controls.

F.4. Official-weight and respondent-equivalent era comparison

Table 28 reports a post-outcome exploratory joint-model comparison of 2023–2024 with 2025–2026. The official-weight row retains the paper’s employment-stock estimand; the respondent-equivalent row is a deliberately different routed-count outcome used only to diagnose whether the later pattern mechanically appears only in final-weight stocks. It cannot recover unpublished age-by-occupation counterfactual weights.

TABLE 28. Post-outcome exploratory comparison of early and later postperiods

Cell construction	2023–2024 coefficient [95% CI]	2025–2026 coefficient [95% CI]	Later minus early [95% CI]	Paired MDE ₈₀
Official final-weight stocks	–0.111 [–0.197, –0.024]	–0.166 [–0.270, –0.062]	–0.056 [–0.123, 0.012]	0.097
Respondent-equivalent routed counts	–0.103 [–0.193, –0.012]	–0.183 [–0.277, –0.089]	–0.080 [–0.139, –0.021]	0.085

Notes: This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper’s employment-stock object. Respondent-equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that population-control revisions caused the later estimate.

G. Linked Flows and Worker Outcomes

G.1. Link construction and risk sets

Person links follow official IPUMS/Census matching guidance and use the available longitudinal weight for each eligible link. Adjacent-month and twelve-month endpoints are distinct samples. A twelve-month link is not the sum of twelve observed monthly transitions and may conceal intervening changes. Table 29 reports fixed-support retention ranges across exposure quintiles rather than collapsing different periods and risk sets into one number.

Employment exit begins with an employed person in an occupation on the analysis support. Reported occupation change requires employment at both endpoints and a valid occupation at each. Entry-destination allocation begins with a nonemployed origin, ends with observed employment, and asks whether the destination falls in Q5 rather than Q1. It therefore conditions on entry and is neither an employment-finding probability nor an employer hiring rate. Every crosswalk descendant of one source record shares its link and resampling weight.

TABLE 29. Official-weight link retention across exposure quintiles

Horizon	Origin age	Preperiod range	Postperiod range
Adjacent month	Ages 22–25	87.6–88.9%	85.3–86.9%
Adjacent month	Ages 26–65	90.5–92.4%	87.6–90.5%
Twelve month	Ages 22–25	50.1–54.0%	48.7–52.8%
Twelve month	Ages 26–65	68.8–73.3%	63.9–70.2%

Notes: Each entry is the minimum–maximum across fixed beta quintiles among employed origins on the 468-occupation analysis support. Numerators require a validated endpoint and a positive horizon-specific longitudinal weight; denominators use origin WTFINL. Rotation ineligibility and a target month outside the extract are recorded separately from eligible origins without a validated endpoint.

Retained and eligible-unlinked origins differ observably. Among postperiod older Q5 origins at twelve months, retained records average age 44.69 and have a 59.25-percent BA-or-higher share, versus 41.50 and 53.35 percent among eligible-unlinked records. Among postperiod young Q2 origins, the 35-or-more hours share is 60.83 percent retained versus 66.24 percent unlinked. These comparisons use origin WTFINL; they diagnose observable selection but do not correct selection or sign bias from unobserved outcomes.

G.2. Results and uncertainty

At the adjacent-month horizon, the young-relative Q5–Q1 estimate is 0.132 for employment exit, 0.003 for reported occupation change, and -0.070 for entry-destination allocation. Their respective 95-percent intervals are $[-0.042, 0.307]$, $[-0.118, 0.123]$, and $[-0.250, 0.110]$. At twelve months, the three estimates are 0.123, -0.018 , and -0.052 . Their intervals are $[-0.111, 0.357]$, $[-0.097, 0.061]$, and $[-0.251, 0.147]$. All include zero. Person- and household-cluster sensitivities, descriptive transition rates, effective risk counts, and named occupation influence appear in the machine-readable flow directory.

Usual-hours changes are estimated within linked employed persons. The adjacent estimate is -0.119 hours per week with interval $[-0.295, 0.057]$. Weekly earnings use the eligible outgoing-rotation sample and its earnings weight; the conditional log-earnings coefficient is -0.0245 with interval $[-0.1024, 0.0534]$. The earnings extract covers 74 months through March 2023 and therefore contains only three postperiod months. Both outcomes condition on employment. Unemployment duration is described only among persons observed unemployed, and no occupation is imputed to never-employed or out-of-labor-force respondents. These restrictions prevent the outcomes from being misreported as population employment effects.

TABLE 30. Employment-exit components in linked CPS records

Endpoint	Component	Raw events	Coefficient	95% interval
Adjacent month	Unemployment entry	32,488	0.055	[−0.272, 0.381]
Adjacent month	Labor-force exit	74,480	0.135	[−0.064, 0.333]
Twelve month	Unemployment at endpoint	27,330	0.246	[−0.295, 0.787]
Twelve month	Out of labor force at endpoint	78,773	0.078	[−0.181, 0.337]

Notes: Rows use official longitudinal weights and the same young-relative Q5–Q1 log-rate-ratio contrast as the core flow table. Counts are repeated link events rather than unique persons. Unemployment and out-of-labor-force events exhaust observed employment exits, but their nonlinear coefficients do not add. Every interval includes zero.

G.3. Selection sensitivity and missing-outcome bounds

Table 31 compares three descriptive linear probability contrasts. Official link weights and origin-weight complete cases use the retained population directly. Observable poststratification instead reweights within fixed observed strata and therefore requires conditional missing-at-random. Factors are neither trimmed nor capped; their maximum is 102.7. Similarity across those three columns cannot rule out selection on unobservables.

TABLE 31. Linked-flow selection sensitivities

Horizon	Margin	Official link	Origin-weight complete case	Observable poststratification	Linear accounting interval
Adjacent	Employment exit	0.0088	0.0088	0.0086	[−0.441, 0.463]
Adjacent	Unemployment entry	0.0037	0.0039	0.0038	[−0.445, 0.459]
Adjacent	Labor-force exit	0.0051	0.0050	0.0048	[−0.444, 0.460]
Twelve month	Employment exit	0.0166	0.0167	0.0210	[−1.570, 1.575]
Twelve month	Unemployment entry	0.0119	0.0128	0.0130	[−1.573, 1.572]
Twelve month	Labor-force exit	0.0047	0.0039	0.0080	[−1.576, 1.569]

Notes: The first three numerical columns are descriptive linear-probability Q5–Q1, young–older, post-pre differences, not nonlinear regression coefficients. Observable poststratification assumes outcome missingness is independent of linkage within fixed observed strata. Accounting intervals impose no conditional-MAR assumption or Lee trimming and may exceed the probability-contrast parameter space because they bound the linear regression contrast jointly over source records.

The last column assigns one common binary missing outcome to each eligible source record and all of its bridge descendants. Its linear interval is sharp under that source-record coupling for the named margin. No conditional-MAR assumption or Lee trimming is used. The analogous log intervals are only conservative marginal outer intervals because bridge coupling need not preserve joint sharpness. These accounting intervals concern missing outcomes, whereas Main Table 5’s confidence intervals concern sampling uncertainty.

G.4. Entry denominators and annual timing

Table 32 places every observed destination on one retained-nonemployment denominator. The destination probabilities sum to one after including people who remain nonemployed and employed destinations outside the exposure support. The conditional Q1–Q5 allocation used in the regression divides only by observed supported entries and remains a different object; neither is employer hiring.

TABLE 32. Employment-entry destinations on a common risk set

Horizon	Age	Period	Remain nonemployed	Q1	Q2	Q3	Q4	Q5	Outside support	Any entry
Adjacent	26–65	Pre	.9154	.0221	.0180	.0117	.0087	.0113	.0128	.0846
Adjacent	26–65	Post	.9179	.0210	.0175	.0115	.0091	.0105	.0125	.0821
Adjacent	22–25	Pre	.8491	.0361	.0398	.0191	.0129	.0211	.0219	.1509
Adjacent	22–25	Post	.8626	.0361	.0352	.0162	.0117	.0193	.0190	.1374
Twelve month	26–65	Pre	.8169	.0439	.0396	.0252	.0213	.0263	.0268	.1831
Twelve month	26–65	Post	.8276	.0399	.0385	.0230	.0215	.0246	.0250	.1724
Twelve month	22–25	Pre	.6408	.0772	.0885	.0492	.0390	.0589	.0465	.3592
Twelve month	22–25	Post	.6611	.0695	.0843	.0555	.0351	.0530	.0415	.3389

Notes: The denominator is all retained nonemployed origins, weighted with the horizon-specific official link weight. Q1–Q5 are employed destinations on the retained exposure support. Outside support records an employed endpoint with a valid occupation outside that support. Any entry equals one minus remaining nonemployment and reconciles to the sum of all observed employed destinations.

The annual exclusion rule drops origins whose twelve-month window straddles January 2023 and is reported with the other primary margins in main Table 5. The exposure-duration sensitivity retains those origins and codes the fraction of the twelve destination months at or after onset. Age and exposure remain defined at the origin. Table 33 reports only this additional sensitivity, avoiding a second finite-draw interval for the primary target. No paired interval comparing the two rules is available, so their difference is not described as detected or equivalent. October 2025 is absent rather than interpolated, and annual endpoints remain distinct from sums of monthly transitions.

G.5. Why no stock–flow calibration is reported

A compatible accounting identity would require entries from nonemployment, exits, switches among all occupations, aging into and out of ages 22–25, corresponding movements for the older denominator, job-to-job moves, and population/sample-composition residuals. The CPS endpoint links do not measure all of these components on a common employer-match risk set, and the BCC hiring and separation objects are not identical to

TABLE 33. Twelve-month exposure-duration sensitivity

Margin	Exposure-duration coding
Employment exit	0.158 [−0.070, 0.387]
Unemployment entry	0.186 [−0.337, 0.708]
Labor-force exit	0.144 [−0.084, 0.372]
Reported occupational outflow	−0.008 [−0.082, 0.067]

Notes: Entries are Q5–Q1, young–older onset coefficients under exposure-duration coding, which codes the share of the twelve destination months at or after onset (102 months). The baseline exclusion rule, which drops origins whose twelve-month link straddles January 2023, is the primary specification and is reported in main Table 5 (89 months). This column is not paired with that table. Intervals use 9,999 occupation wild-score draws.

CPS employment transitions. A single implied stock response would therefore be assumption driven. The revision reports the measurable margins and the precise comparability gap instead of manufacturing a calibration.

H. Payroll Bridge and Exposure-Construction Diagnostics

H.1. BCC version and grouping bridge

The comparison uses the August 12, 2026 version of Brynjolfsson–Chandar–Chen and records the accessed paper, dashboard, and workbook in the evidence ledger. Their reported 19-percent young-worker “kept pace” shortfall is a stock-growth counterfactual in ADP data, not a regression coefficient, employment probability, or hiring rate. Their public dashboard forms equal-occupation exposure groups; later regressions employ other weights and employer controls. Exact dashboard memberships and the treatment of ties are not released.

The CPS bridge therefore applies an equal-occupation, tie-preserving top-two-versus-bottom-three approximation on the present support and labels it as such. The pooled estimate is −0.0720, with interval [−0.1223, −0.0218]. The SOC2-by-calendar-month estimate is −0.0168, with interval [−0.0808, 0.0473]. Their paired movement is 0.0553, with interval [−0.0013, 0.1118]. The intermediate SOC2-by-post specification moves by 0.0573 relative to the pooled model, with interval [0.0005, 0.1141]; this post-outcome bridge contrast is detected narrowly, whereas the stronger family-month comparison is not. The archive provides group summaries and a membership file; it cannot provide concordance with undisclosed assignments. Differences in occupation universe, age comparison, denominator,

endpoint, firm controls, and employer-matched flows remain.

TABLE 34. Post-outcome exploratory BCC-style grouping bridge

Grouping rule	Conditioning structure	Coefficient [95% CI]	Change from baseline [95% CI]	Paired MDE ₈₀
<i>Panel A. Public-dashboard equal-occupation approximation</i>				
Equal occupation	Occupation and calendar-month FE	−0.0720 [−0.1223, −0.0218]	—	—
Equal occupation	SOC2 by young by post	−0.0148 [−0.0792, 0.0497]	0.0573 [0.0005, 0.1141]	0.083
Equal occupation	SOC2 by young by calendar month	−0.0168 [−0.0808, 0.0473]	0.0553 [−0.0013, 0.1118]	0.082
<i>Panel B. Historical employment-weighted approximation</i>				
Employment weighted	Occupation and calendar-month FE	−0.075 [−0.125, −0.025]	—	—
Employment weighted	SOC2 by young by post	−0.027 [−0.095, 0.041]	0.048 [−0.011, 0.108]	0.086
Employment weighted	SOC2 by young by calendar month	−0.027 [−0.093, 0.040]	0.048 [−0.011, 0.107]	0.086

Notes: All rows are post-outcome exploratory CPS bridges, not replications of BCC’s proprietary ADP analysis. Each row compares GPT-4-beta Q4–Q5 with Q1–Q3 on the same 468 occupations, includes the standardized historical Webb-software control, and uses 9,999 common occupation-level Rademacher multipliers. Paired changes compare each SOC2-conditioned row with its own unconditioned baseline. Panel A follows the public dashboard’s equal-occupation description as closely as public information permits. Panel B uses earlier employment-weighted cuts formed from 108 months of young-plus-older CPS stock, including postperiod months, and is only a grouping-rule diagnostic. Exact BCC occupation membership, universe, cutoff and tie algorithm are not public, so concordance is unverified. The BCC and CPS objects also differ in ages, denominator, endpoint, employer controls, and observed worker–firm flows. Null-containing paired intervals are nondetections, not equivalence.

H.2. Annual ACS reconstruction

The annual extension uses ordinary one-year ACS PUMS for 2017–2019 and 2021–2024; no standard 2020 one-year file exists, and no experimental 2020 estimate is spliced into the panel. Eighty replicate weights per endpoint year form the successive-difference-replication intervals in Table 35. The BCC-labelled rows remain public analogues: complete BCC occupation membership is unavailable, so they cannot be exact replications.

For the benchmark, changing from all employed to full-time civilian wage-and-salary workers is not detected under common ACS replicate perturbations: the paired differences are −0.0025 ([−0.0266, 0.0216]) for the equal-occupation BCC analogue and 0.0056 ([−0.0174, 0.0285]) for fixed primary groups. The fixed-primary versus BCC-analogue grouping difference is likewise not detected among all employed workers (0.0090, [−0.0045, 0.0224]). Among full-time civilian wage-and-salary workers that grouping difference is narrowly detected (0.0171, [0.0002, 0.0340]), which reinforces rather than resolves the missing-membership limitation.

For the annual young-relative regressions, full-calendar family-year estimates retain negative ACS sampling intervals under the fixed primary exposure groups, while the broader equal-occupation grouping and every nonreuse-calendar family-year interval

TABLE 35. Independent annual ACS reconstructions

Definition	Population	Calendar/model	Estimate [ACS CI]	Family-year [ACS CI]	Family-year occupation CI	Family-year family CI
<i>Panel A. 2022–2024 Q5–Q1 growth-factor benchmark, ages 22–25</i>						
BCC published	All employed	2022–2024	–0.022 [–0.055, 0.011]	–	–	–
ACS analogue, equal occupation	All employed	2022–2024	–0.0859 [–0.1238, –0.0480]	–	–	–
Aligned annual CPS analogue	All employed	2022–2024	–0.0346 [–0.1455, 0.0763]	–	–	–
BCC published	Full-time civilian wage/salary	2022–2024	–0.019 [–0.059, 0.020]	–	–	–
ACS analogue, equal occupation	Full-time civilian wage/salary	2022–2024	–0.0834 [–0.1272, –0.0396]	–	–	–
Aligned annual CPS analogue	Full-time civilian wage/salary	2022–2024	–0.0306 [–0.1366, 0.0755]	–	–	–
<i>Panel B. Annual young-relative Q5-by-post regressions, ages 22–25 versus 26–65</i>						
Fixed primary groups	All employed	Full	–0.1089 [–0.1310, –0.0869]	–0.0474 [–0.0885, –0.0063]	[–0.1316, 0.0368]	[–0.1242, 0.0294]
Fixed primary groups	All employed	Nonreuse	–0.0951 [–0.1204, –0.0698]	–0.0220 [–0.0705, 0.0264]	[–0.1121, 0.0680]	[–0.1013, 0.0572]
Fixed primary groups	Full-time civilian wage/salary	Full	–0.1196 [–0.1456, –0.0936]	–0.0691 [–0.1193, –0.0188]	[–0.1554, 0.0173]	[–0.1777, 0.0395]
Fixed primary groups	Full-time civilian wage/salary	Nonreuse	–0.1063 [–0.1361, –0.0764]	–0.0467 [–0.1057, 0.0122]	[–0.1317, 0.0383]	[–0.1480, 0.0546]
Broader equal-occupation groups	All employed	Full	–0.0830 [–0.1053, –0.0607]	–0.0151 [–0.0559, 0.0257]	[–0.1000, 0.0698]	[–0.0942, 0.0639]
Broader equal-occupation groups	All employed	Nonreuse	–0.0723 [–0.0979, –0.0467]	–0.0056 [–0.0535, 0.0424]	[–0.0918, 0.0806]	[–0.0927, 0.0815]
Broader equal-occupation groups	Full-time civilian wage/salary	Full	–0.0933 [–0.1203, –0.0664]	–0.0292 [–0.0790, 0.0205]	[–0.1112, 0.0527]	[–0.1307, 0.0722]
Broader equal-occupation groups	Full-time civilian wage/salary	Nonreuse	–0.0828 [–0.1137, –0.0520]	–0.0291 [–0.0875, 0.0293]	[–0.1118, 0.0535]	[–0.1294, 0.0712]

Notes: Panel A reports growth-factor differences without an older comparison group. The public analogues do not recover BCC’s undisclosed complete occupation membership. Panel B reports log coefficients. Full uses 2017–2019 and 2021 as preperiod years and 2023–2024 as postperiod years; nonreuse uses 2017 and 2021 as preperiod years. Standard 2020 one-year ACS PUMS are omitted, and 2022 is the transition year. ACS intervals use successive-difference replication with 80 replicate weights per perturbed endpoint year and a year-block independence approximation. Occupation and family intervals instead assign shocks at those cluster levels; the procedures answer different stochastic questions and are not combined. All extensions are post-outcome descriptive analyses.

contain zero. The paired ACS sampling intervals for the pooled-minus-family-year movements exclude zero in all eight reported definition–population–calendar combinations. This is not universal across stochastic targets: supplementary occupation- and family-multiplier intervals are wider, and every family-year coefficient reported in Table 35 includes zero under those shock assignments. The annual exercise thus shows that large respondent counts sharpen person-sampling uncertainty but do not create within-family occupational comparisons or identify an AI effect.

H.3. Equal-occupation estimation objective

The public-dashboard grouping bridge above changes the grouping rule. A separate sensitivity changes the regression objective itself: each occupation receives total weight one, while WTFINL still determines the young and older shares within occupation–month cells. Under this objective the pooled Q5-by-post coefficient is –0.0833, compared with –0.1321 under employment-stock weighting. Their paired difference is 0.0488 ([–0.0604, 0.1581]). The corresponding family-month estimates are –0.0047 and –0.0217, with a paired difference of 0.0170 ([–0.1587, 0.1927]). The design therefore does not detect an objective-driven difference, but the wider equal-occupation intervals and null-containing paired intervals do not establish equivalence.

TABLE 36. Equal-occupation objective sensitivity

Conditioning structure	Objective	Q5 × post [95% CI]	Equal occupation – stock weighted [95% CI]
Occupation and calendar-month FE	Employment-stock weighted	−0.132 [−0.220, −0.045]	—
Occupation and calendar-month FE	Equal occupation	−0.083 [−0.206, 0.040]	0.049 [−0.060, 0.158]
SOC2 family by calendar month	Employment-stock weighted	−0.022 [−0.159, 0.115]	—
SOC2 family by calendar month	Equal occupation	−0.005 [−0.203, 0.193]	0.017 [−0.159, 0.193]

Notes: This sensitivity changes the estimation objective: every occupation receives total weight one, while WTFINL continues to determine young and older shares within occupation–month cells. Treatment membership, 468-occupation support, calendar, and nuisance structures are unchanged. Intervals use 9,999 common occupation-level Rademacher score multipliers; paired differences are formed before applying the common draws. A paired interval containing zero means that this design does not detect a difference, not that the objectives are economically equivalent.

H.4. Task-capability weighting

Write the two Eloundou primitives as E_1 (the E1 share) and E_2 (the E2 share). The score family $E_1 + \lambda E_2$ is evaluated on a fixed grid from zero to one under common preperiod weights and common bootstrap draws. The implementation verifies numerically that $\lambda = 0.5$ reproduces raw beta before grouping. Any earlier mismatch between that row and the headline arose from a different support or treatment contract and is preserved only as implementation history.

Two versions are kept separate. The native-rank analysis recomputes the score normalization and tie-preserving groups at each λ , so both scale and membership can change. The fixed-membership analysis changes only the continuous score representation while holding the beta groups fixed. For every grid point the result files report coefficient, paired difference from beta, interval, occupation coverage, tail turnover, and correlations with computer use and remotability. A focused 408-occupation run adds both computer use and remotability on common support. None of these paths measures realized deployment.

H.5. Primitive and architecture comparisons

The joint primitive model reports E_1 and E_2 in raw share units, in preperiod-standard-deviation units, their covariance, and an illustrative common change. This is a two-regressor parameterization, not a causal decomposition into technologies. Architecture comparisons likewise report both component coefficients and their paired difference on each declared support. Webb-AI comparisons are shown first on identical support and then on broader native support so that score changes are not confused with population changes. Coordinate rotations that do not change the fitted predictor are not interpreted

as additional scientific results.

Raw AIOE is a standardized source-occupation index: one point is approximately one unweighted standard deviation across the 774 occupations in the source workbook, and the source interquartile difference is 1.879 points. It is not a percentage. Equal-administrative and source-employment-weighted mappings preserve that source scale through averaging, although their target-support distributions need not retain unit variance. The direct-ability reconstruction has its own raw scale and is not numerically interchangeable. Categorical AIOE estimates use ranks and tie-preserving groups; continuous comparisons must state whether their increment is a raw point or a current-support weighted standard deviation.

Table 37 runs the five non- λ definitions on the current 113-month outcome contract. Native-support pooled coefficients range from -0.0991 for direct-ability AIOE to -0.0145 for the reversed OECD gap. Every native-support alternative-minus-beta paired occupation interval contains zero; the OECD comparison nearly reaches the boundary, with interval $[-0.0004, 0.2378]$. On the common 444-occupation AIOE support, however, family-month differences between administrative/equal-component mapping and direct ability (0.0688 , $[-0.0070, 0.1447]$ under occupation clustering but $[0.0057, 0.1320]$ under family clustering) and source-employment weighting (0.0586 , $[0.0111, 0.1061]$ under occupation clustering) show that implementation can matter for the conditioned result. These are paired implementation sensitivities within AIOE, not independent replications or causal technology decompositions.

TABLE 37. Non- λ exposure definitions on the current CPS contract

Exposure definition	Occupations	Pooled coefficient [95% CI]	Alternative – beta [95% CI]	Family-month coefficient [95% CI]	Alternative – beta [95% CI]
AIOE, administrative/equal component	466	$-0.072 [-0.143, -0.001]$	$0.064 [-0.025, 0.152]$	$0.065 [-0.050, 0.181]$	$0.095 [-0.063, 0.253]$
AIOE, direct ability	455	$-0.099 [-0.174, -0.024]$	$0.027 [-0.059, 0.113]$	$0.024 [-0.084, 0.132]$	$0.011 [-0.143, 0.165]$
AIOE, source-employment weighted	456	$-0.072 [-0.145, 0.000]$	$0.068 [-0.020, 0.157]$	$0.071 [-0.044, 0.186]$	$0.114 [-0.047, 0.274]$
Webb artificial intelligence	443	$-0.068 [-0.160, 0.023]$	$0.067 [-0.036, 0.171]$	$-0.079 [-0.177, 0.019]$	$-0.036 [-0.203, 0.131]$
OECD reversed AI-skills gap	448	$-0.014 [-0.096, 0.067]$	$0.119 [-0.0004, 0.238]$	$0.069 [-0.033, 0.171]$	$0.065 [-0.125, 0.255]$

Notes: Rows use each alternative’s native finite-score support. The outcome cells, age contrast, calendar, estimator, Webb-software control, and preperiod weighting rule are fixed. Each beta anchor is rebuilt on exactly the alternative row’s support. Coefficients compare Q5 with Q1; intervals and paired differences use 9,999 common occupation-level Rademacher score multipliers. A null-containing paired interval is a nondetection, not equivalence. Continuous-score estimates and raw-to-weighted-SD scaling are preserved in the public result files.

H.6. Later self-reported use

An external descriptive check merges the frozen beta score to the author-maintained Real-Time Population Survey occupation workbook. The workbook pools four waves from Au-

TABLE 38. Post-outcome exploratory direct and software-complement primitives

Units or contrast	Term	Estimate [95% CI]	Bootstrap SE	MDE ₈₀
Raw share	$E_1 \times \text{young} \times \text{post}$	-0.189 [-0.354, -0.024]	0.085	0.239
Raw share	$E_2 \times \text{young} \times \text{post}$	-0.089 [-0.172, -0.005]	0.043	0.121
Preperiod SD	$E_1 \times \text{young} \times \text{post}$	-0.031 [-0.058, -0.004]	0.014	0.039
Preperiod SD	$E_2 \times \text{young} \times \text{post}$	-0.028 [-0.054, -0.002]	0.014	0.038
One SD in each primitive	$E_1 + E_2$ joint change	-0.059 [-0.098, -0.020]	0.020	0.056

Notes: All entries are post-outcome exploratory and use the same 468 occupations and 9,999 common occupation multipliers. The joint model includes E_1 , E_2 , and standardized Webb software, each interacted with $\text{young} \times \text{post}$. Here E_1 is the share of tasks classified as directly accelerable and E_2 is the share requiring complementary software; beta satisfies $E_1 + 0.5E_2$. The preperiod employment-weighted means (standard deviations) are 0.144 (0.164) for E_1 and 0.463 (0.315) for E_2 . The common-draw correlation between the two standardized coefficient draws is 0.062. Component-row MDEs use the analytic occupation-clustered standard errors; the joint-change MDE uses its common-draw standard error. The final row is the declared one-standard-deviation increase in both primitives, not a paired difference between two technologies. The parameterization does not measure adoption and is not a causal decomposition.

TABLE 39. Post-outcome exploratory task-weight architectures on common support

Architecture pair	Alternative coefficient	Beta coefficient	Alternative – beta [95% CI]	Paired MDE ₈₀
E_1 versus $E_1 + 0.5E_2$	-0.103	-0.132	0.029 [-0.044, 0.103]	0.106
$E_1 + 0.25E_2$ versus $E_1 + 0.5E_2$	-0.101	-0.132	0.031 [-0.005, 0.068]	0.055
$E_1 + 0.75E_2$ versus $E_1 + 0.5E_2$	-0.143	-0.132	-0.011 [-0.039, 0.017]	0.041
$E_1 + E_2$ versus $E_1 + 0.5E_2$	-0.156	-0.132	-0.024 [-0.064, 0.017]	0.058

Notes: All entries are post-outcome exploratory Q5–Q1 coefficients estimated on the identical 468-occupation support. Beta is $E_1 + 0.5E_2$. At each alternative weight, normalization, cutoffs, and tie-preserving quintile memberships are recomputed, so a paired movement combines score and membership changes rather than holding the treatment groups fixed. The intervals use 9,999 common occupation multiplier draws, preserving covariance within each pair. Every paired interval contains zero; this means that the design does not detect a difference, not that the architectures are economically equivalent.

TABLE 40. Post-outcome exploratory Webb conditioning and support

Specification	Occupations	Beta coefficient [95% CI]	Same-support paired change [95% CI]
Without Webb, fixed primary support	468	-0.134 [-0.219, -0.049]	-0.002 [-0.010, 0.007]
With Webb, fixed primary support	468	-0.132 [-0.220, -0.044]	Reference
Without Webb, broader beta-valid support	490	-0.135 [-0.218, -0.051]	Not paired

Notes: All entries are post-outcome exploratory. The first two rows hold occupations, beta cutoffs, and the outcome specification fixed; the displayed paired change is without-Webb minus with-Webb, uses common draws, and has $\text{MDE}_{80} = 0.013$. Its interval containing zero is a nondetection, not equivalence. The final row relaxes Webb availability and therefore changes support and beta cutoffs; it has no paired interpretation. Webb conditioning is an occupational-software proxy, not realized software or AI adoption.

gust 2025 through May 2026; it is not a preperiod measure or a detailed occupation-by-time panel. The exact-code merge retains 121 nonsuppressed occupations from the frozen 468, with all five quintiles and 20 major families represented. The unweighted exposure-adoption correlation is 0.504 and mean adoption differs by 0.267 between Q5 and Q1, although the quintile profile is not monotone because Q4's mean exceeds Q5's. After major-family means are removed, 26.2 percent of the exposure sum of squares remains and the residualized slope is 0.302.

The workbook supplies pooled point estimates and respondent counts but no detailed-cell standard errors or replicate weights. Table 41 therefore reports no p-values or confidence intervals, and its respondent-count weighting is only a descriptive sensitivity. These results show positive alignment between the capability score and later reported use; they do not establish adoption timing, explain the CPS coefficient, or identify an employment effect.

TABLE 41. Later self-reported GenAI use by frozen beta exposure

	Equal occupation	Respondent-count sensitivity
Pearson correlation	0.504	0.647
Spearman correlation	0.494	—
Adoption-on-beta slope	0.493	0.695
Mean adoption, Q1	0.263	0.250
Mean adoption, Q5	0.530	0.583
Q5 minus Q1	0.267	0.333
Within-family exposure SS share	0.262	0.191
Within-family slope	0.302	0.465

Notes: The exact-code merge retains 121 nonsuppressed occupations from the frozen 468; all five quintiles and 20 SOC major families are represented. The RPS workbook pools four waves from August 2025 through May 2026 and suppresses rates based on fewer than 20 respondents. The second column weights occupations by the workbook's pooled unweighted respondent count, which is neither a survey weight nor a verified inverse-variance weight. Detailed-cell standard errors and replicate weights are unavailable, so the table reports no inference. The quantities are descriptive validation of score alignment, not an adoption treatment or an explanation of the CPS coefficient.

H.7. What validation would require

None of the sampling intervals above treats a score as a noisy observation of a validated true exposure. A useful validation sample would begin at the versioned occupation-task level and oversample both broad families and tasks on which architectures disagree. It

would record separate outcomes for technical capability, realized time savings without complementary software, realized time savings with software, observed workplace adoption, and task importance. Conflating these labels would reproduce the construct problem the validation is meant to solve.

Labels should come from repeated, blinded, independent expert assessments and, where feasible, audited product-use or task-performance records. Calibration and evaluation occupations should be separated so that crosswalk and occupation aggregation choices are tested out of sample. The task description, model snapshot, prompt, complementary tools, rater instructions, occupational bridge, and employment weights all require version records. Sample size cannot be stated as one universal number: it depends on within-occupation task correlation, rater reliability, unequal task weights, the latent quantity being corrected, and whether the downstream target is a continuous slope or tail assignment. In the absence of these inputs, the revision specifies the validation design but does not apply a generated-regressor correction.

I. Reproducibility, Numerical Consistency, and Unresolved Items

I.1. Execution contract

The revision archive separates restricted inputs from distributable code and derived aggregate outputs. Two commands serve different purposes. The restricted-data orchestrator authenticates the licensed inputs, refuses to overwrite an existing output root, runs the rebuilt baseline before dependent modules, passes named treatment objects downstream, invokes each self-check, and runs the pinned official HonestDiD implementation. A separate aggregate-only command regenerates exhibits from committed results, checks numerical and prose consistency, runs repository tests, and builds the LaTeX documents. The latter neither reads CPS microdata nor independently re-estimates the empirical models. Environment records report Python and R package versions, seeds, scheduler resources, source hashes, and git state where captured. Every module declares its treatment contract, calendar, support, objective, inference target, and output paths before interpreting the result.

The retained R3 `baseline_inventory` directory contains a provisional machine-readable allowlist (`PUBLIC_REPLICATION_MANIFEST.csv`) and a written scan contract (`INPUT_PROVENANCE_RECEIPT.md`). They have not yet been updated and applied to a separately staged V3 release candidate. Accordingly, neither the current Git check-out nor this delivery is labeled a sanitized public-release artifact. Before release, a new

empty stage must be built from the updated positive allowlist and actually checked for private absolute roots, credential-token patterns, restricted-data formats, symlinks, caches, and direct CPS link-identifier columns, followed by manual disclosure, licensing, and PDF-metadata review. Licensed CPS records, account metadata, credentials, longitudinal identifiers, and operational path history are not eligible for that future stage. The retained input manifest records IPUMS extract numbers 9, 10, and 11 and available request, data, DDI, and codebook hashes; unavailable API timestamps remain blank rather than inferred.

The canonical results ledger is the numerical source of truth. Manuscript tables are generated from or checked against it, and repeated estimates use one declared draw set. If an interval comes from a different stochastic target—occupation shocks, broad-family shocks, or household resampling—the table labels that procedure rather than presenting multiple intervals as interchangeable versions of one result. The test suite verifies calendar membership, one-sided zeros, route conservation, fixed-support pairing, covariance symmetry and rank, draw hashes, and table/prose consistency.

I.2. Implemented failures

The failure registry is part of the replication evidence. It includes the literal vintage merge, the original March sample selection, nonconvergence of saturated seasonality variants, and indefinite positive-lag joint HAC matrices from the superseded historical-treatment audit. A failure is never converted into a different estimator without an explicit amendment. Successful reruns preserve the earlier log and record the replacement’s package and solver versions; operational installation history remains outside the scientific exposition.

I.3. Unresolved limitations

The analysis cannot observe firm adoption, employer identifiers, worker–firm hiring, confidential CPS design variables, exact BCC dashboard memberships, age-specific occupational route probabilities, or a validated latent “true” AI exposure. The public data therefore do not identify an AI causal effect or a complete stock–flow mechanism. The direct within-family tail population is small, unrestricted preperiod differences are present, several linked-flow intervals are wide, and January 2025–2026 survey-production changes cannot be undone with published aggregate factors. The annual ACS extension omits the unavailable standard 2020 one-year file, combines year-specific sampling components under a block-independence approximation, and cannot recreate BCC’s undisclosed membership or missing family-by-tail support. These are limits on interpretation, not missing

robustness rows that a specification search can solve.

Author affiliation, email, acknowledgments, funding, conflicts, data-access wording, and journal-required disclosures remain marked [AUTHOR TO COMPLETE] wherever author-supplied information is unavailable. No identifying or disclosure information is inferred from directory names, credentials, or institutional access.

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